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MILITARY SPECIFICATION

CONTROL ASSEMBLY, ATTITUDE, GYROSCOPIC AF/A24G-1A

1. SCOPE

1.1 This specification covers the requirements for one type of gyroscope attitude control assembly, designated AF/A24G-1A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-C-5015	Connectors, Electrical, "AN" Type
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-P-7788	Plate, Plastic, Lighting
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-F-15160	Fuse, Style FO3
MIL-S-16892	Synchros, 400 Cycles
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment
MIL-D-25521	Detector, Magnetic, Azimuth DT-173/AJN
MIL-C-26601	Controller, Third Gimbal DCK-10/A24G-1

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage

FSC 6615

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MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MS17322	Meter, Time Totalizing Miniature, Digital 115 Volt 400 Cycle
MS24534	Meters, Time Totalizing, Round (Indicator, Elapsed Time)
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33586	Metals, Definition of Dissimilar

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The control assembly shall consist of the following components:

a. Gyroscope assembly, displacement SBK-8/A24G-1A

b. Amplifier-power supply ASK-12A/A24G-1A

c. Adapter, compass ADK-182A/A24G-1A

d. Controller, compass DCK-100/A24G-1A.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent. If used within a hermetically sealed enclosure, fungicidal treatment shall not be required.

3.4.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to conditions likely to be encountered in storage or normal service.

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3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the control assembly, except where magnetic materials are essential.

3.4.4 Corrosive fumes. Materials, as installed and used in the control assembly and under the service conditions specified in section 4, shall not liberate deleterious fumes.

3.4.5 Protective treatment. When materials are used in the construction of the control assembly that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction. The control assembly components shall be designed and constructed to meet the requirements specified herein and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. Corresponding components of all control assemblies procured shall be interchangeable. No gimbal binding under differential pressure conditions shall occur as a result of interference between case and gimbal should a leak develop in service.

3.5.1 The components shall be designed to accept and provide input and output signals as specified in section 4.

3.5.2 Reliability. The control assembly shall have a minimum acceptable mean-time-between-failures (MTBF) of 300 hours. The reliability test program shall be based upon a discrimination ratio of 2.0 and a confidence level of 90 percent.

3.5.2.1 Longevity. The control assembly shall have an operating life span equipment longevity as defined in MIL-STD-781 of not less than 2,000 hours before it consistently fails to meet the MTBF index.

3.6 Performance. The control assembly shall be capable of operating as specified herein under the following conditions:

a. Temperatures ranging from -54° to +98°C

b. Dielectric strength of 500V dc with a resistance of not less than 50 megohms

c. Vibration up to 10g

d. Acceleration up to 20g

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e. Altitude pressure equivalent to 80,000 feet

f. Shock forces of 15g for 11 milliseconds (basic design) and 30g for 11 milliseconds (crash safety)

g. Humidity up to and including 95 percent

h. Exposure to fungus enzymes

i. Exposure to a simulated rainfall of 5 inches per hour

j. Salt fog for a period of 48 hours

k. Exposure to blowing fine sand and dust particles

l. Operation in explosive atmosphere

m. Exposure to acoustic sound fields that are characteristic of aircraft

n. Sealing thermal shock consisting of 8 cycles in water at 85° ±5°C and then in water at 5° ±4°C

o. Immersion in water with a pressure of 2-1/2 inches Hg applied for 1 minute

p. Exposure to sudden temperature changes.

3.6.1 Radio noise interference and susceptibility. The control assembly shall meet the radio noise interference and susceptibility requirements of MIL-I-6181.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.8 Displacement gyroscope assembly. The displacement gyroscope assembly shall consist of a vertical and a directional gyro mounted in multiple gimbals with a common outer gimbal as described in the following paragraphs. This gimbal arrangement is used in order to avoid the gimbal lock position. The outline dimensions of the reference shall be in accordance with figure 1.

3.8.1 Vertical gyro. The vertical gyro shall have sufficient angular momentum and be of such design so as to meet the performance specified herein when operated throughout the temperature, vibration, and acceleration ranges.

3.8.1.1 Gimbal system. The vertical gyro shall be suspended in a gimbal system having two free inner gimbals and a servo-driven outer gimbal to eliminate gimbal lock of the gyro. The outer gimbal shall be used in common with the directional gyro.

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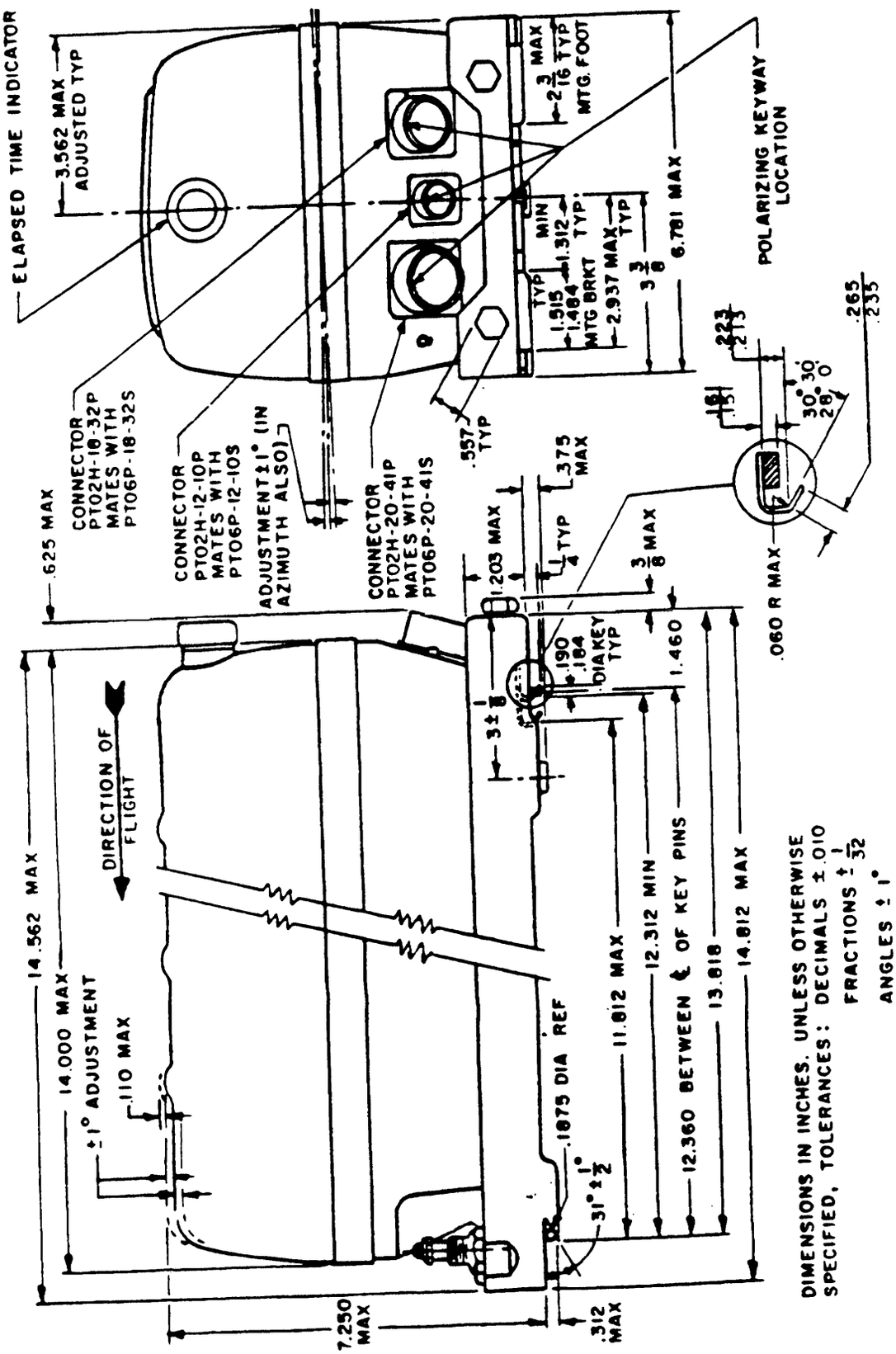


FIGURE 1. Displacement Gyroscope Assembly

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3.8.1.1.1 First vertical gyro gimbal. The first gimbal shall be the gyro rotor housing. It shall be stabilized by the inertia of the gyro.

3.8.1.1.2 Second vertical gyro gimbal. The second gimbal shall be the next gimbal outward from the first gimbal. It shall be stabilized in pitch by the gyro.

3.8.1.1.3 Third vertical gyro gimbal. The third gimbal shall be the outer gimbal. It shall be servo driven to maintain a fixed relation between the first and second gimbals except for certain conditions specified herein.

3.8.1.2 Vertical gyro axes of freedom. The vertical gyro shall have three axes of freedom. Each shall be a signal axis.

3.8.1.2.1 First vertical gyro axis of freedom. The first axis of freedom shall be between the first and second gimbals. This axis shall be maintained in the horizontal plane and be aligned with the orthogonal horizontal projection of the fore-and-aft axis of the reference assembly under normal conditions. A minimum freedom of ±80° shall be provided about this axis. The angular displacement about the axis shall provide an indication which shall change from roll to yaw as the pitch angle changes from 0° to ±90° pitch. Stops shall be incorporated on this axis which shall produce a pitch precession when the freedom is exceeded. No damage shall result under this condition.

3.8.1.2.2 Second vertical gyro axis of freedom. The second axis of freedom shall be between the second and third (servo driven) gimbals. It shall be perpendicular to the first axis of freedom and shall be maintained in the horizontal plane under normal conditions. Unlimited freedom shall be provided about this axis. The angular displacement about the axis shall be a measurement of the pitch angle.

3.8.1.2.3 Third vertical gyro axis of freedom. The third axis of freedom shall be between the third gimbal and the instrument frame or case. This axis shall be perpendicular to the second axis of freedom and shall coincide with the fore-and-aft axis of the instrument. Unlimited freedom shall be provided about this axis. The angular displacement about the axis shall be a measurement of the roll angle.

3.8.1.3 Gimbal actuation. The gyro erection system shall maintain the inner two gimbals in a fixed relation to the apparent gravity vertical. The outer gimbal shall be actuated by a servo mechanism and so driven as to maintain the spin axis of the vertical gyro and the pitch axis in a 90° relation to each other, except under certain transient conditions. The response of the servo of the outer gimbal shall be such that the 90° relation will be maintained during roll rates from zero through 360° per second with no lag, overshooting, oscillating, hunting, or jumping of such a magnitude and frequency which will affect the performance of a high performance autopilot. The servo loop shall be tight enough so that during loop maneuvers, when a third gimbal controller (DCK-10/A24G-1) is not connected, the third gimbal will rotate 180° to keep proper sensing of the servo drive when the third gimbal axis passes through the vertical.

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3.8.1.3.1 Special third gimbal operation. The design of the circuitry of the third gimbal servo shall be such that by external connections, the third gimbal will perform as required under 3.8.1.3, or by externally connecting it to the DCK-10/A24G-1 third gimbal controller, the third gimbal shall be so stabilized that it will not rotate about its axis as it passes through or near the vertical. When, while thus controlled, the reference is rotated in pitch so that the third gimbal axis passes through or near the vertical but remains within ±20° of a vertical plane which is parallel to the heading held at the time the operation was initiated, the third gimbal shall remain stable and shall keep the pitch axis in a plane which is perpendicular to this vertical plane and continuously parallel to the third gimbal axis. The pitch gimbal shall pass through the third gimbal and provide a continuous pitch signal. Displacement of the reference about the third gimbal shall provide a signal of the roll motion throughout the loop or Immelman maneuver. The design shall permit the third gimbal controller to perform the functions required to accomplish the operations specified herein and in MIL-C-26601.

3.8.1.4 Erection. The erection mechanism shall incorporate separate systems for pitch and bank. Erection shall be accomplished by electrolytic switches and torque motors on the gimbals. When the vertical gyro is displaced in one axis, it shall not deviate in the other axis while it is erecting by more than 1.0°. The erection system shall provide both initial and final erection and shall erect the gyro from any attitude to the vertical as determined by the vertical sensing electrolytic switches. The pitch and roll erection torquer leads shall be brought out of the case so that precision voltage can be applied externally to precess the gyro for test purposes.

3.8.1.4.1 Initial erection. The vertical gyro shall have fast initial erection so that it assumes a position within 1° of vertical in pitch and bank within 1-1/2 minutes, or less, after power is applied. The rotor shall be at a standstill and the spin axis may be in any possible attitude at the time the power is applied with or without the third gimbal controller connected.

3.8.1.4.2 Final erection. The vertical gyro shall align its spin axis with the direction of gravity within ±1/4° in 2 minutes, or less, after the initial erection period.

3.8.1.4.3 Normal erection rate. The rate of normal erection about the roll axis shall be nominally 1.3° per minute and shall fall between the limits plotted on figure 2 throughout all environments and roll attitudes of the control case. The rate of normal erection about the pitch axis shall be nominally 1.3° per minute and shall fall between the limits plotted on figure 2 throughout all environments. Pitch erection sensing shall be maintained when displacing the gyro in roll ±45° or less.

3.8.1.4.4 Fast erection. Means shall be provided for fast erecting of the gyro to the apparent vertical of both axes of the gyro simultaneously. This fast erection shall be obtained by externally closing the circuit between pins U and V of table I. The fast erection rate shall be not less than 15° per minute in either axis at normal temperature or 13° per minute at temperature extremes.

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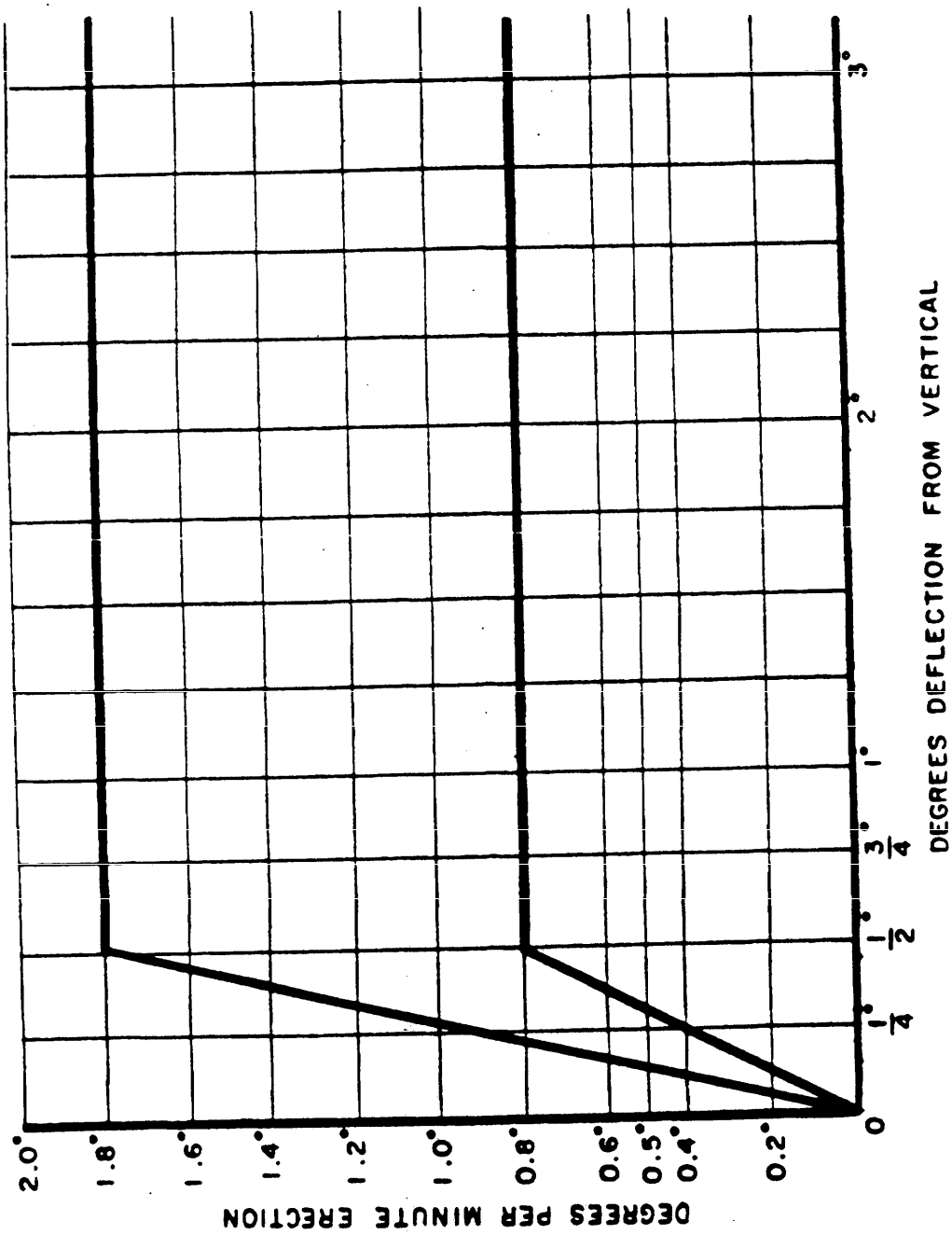


FIGURE 2. Erection Rate As a Function of Deflection

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TABLE I. Amplifier-Power Supply Input Connector (J202)

Bendix P/N PT00P-14-19P, or equal, to mate with Bendix P/N PT06P-14-19S, or equal	
Pin	Function
A (Neutral)	115/200V ac, 400 Hz
B (Phase A)	3-phase, 4-wire WYE
C (Phase B)	Phase rotation A-B-C
D (Phase C)	
E	Roll alarm - to F
F	Roll alarm - to E
G	Spare
H	Reserved
J	Reserved
K	Failure warning
L	Failure warning
M	Pitch erect cutout
N	Roll erect cutout
P	Cutout common - 115V 400 Hz
R	Start 60 sec auto sync - to S
S	Start 60 sec auto sync - to R
T	Reserved
U	Fast erection - to V
V	Fast erection - to U

3.8.1.4.5 Roll and pitch pendulum erection cutout. The design shall be such that when pins N and P of table I are connected externally, roll erection shall be cut out, and when pins M and P of table I are connected externally, pitch erection will be cut out. The current drawn from pin N to pin P and pin M to pin P of table I shall not exceed 125 ma each. The design shall be such that when all three pins are externally commoned, total pendulum erection cutout will occur. Pendulum erection cutout shall not affect the torquing of the gyro resulting from input of the earth's profile and earth rate correction.

3.8.1.4.6 Fast precession. For test purposes, a means shall be provided to apply and remove, from an external source, a fast precession rate in either direction to the pitch axis of the vertical gyro after the gyro is up to speed. This precession rate shall be not less than 20° per minute.

3.8.1.4.7 Vertical gyro correction. Provisions shall be made so that torques may be applied in pitch and roll in either direction. The design shall be such that the torquers will work directly from either an amplifier or a transformer. The torquers shall be internally excited so that when approximately 15V ac, 400 Hz, phase C +90° is

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applied to the control phase legs, the torquer will cause the gyro to precess at a nominal rate of approximately 60° per hour. The control phase impedance shall be approximately 355 ±J260 ohms. The power consumption in each torquer fixed phase shall not exceed 1w.

3.8.1.5 Vertical gyro synchro outputs. Two transmitting synchros shall be mounted on each vertical gyro axis of freedom except the inner roll which shall have one. The synchros shall provide displacement signals of gimbal movement about their respective axes at a 1-to-1 ratio. One each of the synchros on the pitch and outer roll axes shall have their stator and rotor leads brought out to the signal electrical connector (see table II), separate from each other and other circuitry. The synchros shall each be capable of operating up to eight high impedance control transformers, Eclipse-Pioneer AY500-5 type, or equal, when the control transformers are loaded with 10,000-ohm loads. The synchros shall have a nominal no-load line-to-line output of 11.8 ±0.25V and a phase shift of 4.0° ±1.5°. When loaded with 8 CT's, each having a 10,000-ohm load, the line-to-line output voltage shall be not less than 10.6V and the phase shift shall be not more than 10°. The remaining synchros on the vertical gyro axes of freedom shall be internally excited with the voltage across terminals A and D of the control electrical connector (see table III). The stator leads of the synchros shall be brought out through the output connector terminals in accordance with table II; however, they may be connected to other control circuitry provided up to six high-impedance control transformers of the Eclipse-Pioneer AY500-5 type, or equal, loaded with 10,000 ohms can be externally operated from each.

3.8.1.5.1 Synchro zeroing and sensing. The transmitting synchros on the vertical gyro axes of freedom shall be zeroed and their lead terminals shown in table II connected to satisfy the following procedure for identification of the pins.

3.8.1.5.1.1 Externally excited pitch synchro. Pins c and K of table II shall be connected to power lead pin A of table III and pin L of table II shall be connected to power lead pin D of table III. The voltage across pins a and b and the voltage across pins L and a or b of table II shall be measured with the gimbals at zero displacement. The voltage across pins a and b of table II shall be at the minimum and the voltage across L and a or b of table II shall be less than the voltage applied to pins A and D of table III. The voltage across pins a and c and across b and c of table II shall be measured as the reference assembly is rotated in climb about the pitch gimbal. The voltage across pins a and c of table II shall increase before decreasing and the voltage across pins b and c of table II shall decrease before increasing.

3.8.1.5.1.2 Externally excited outer roll synchro. Pins W and H of table II shall be connected to power lead pin A of table III, and pin J of table II shall be connected to power lead pin D of table III. The voltage across pins U and V of table II and the voltage across pins J and U or V of table II shall be measured with the gimbal at zero displacement. The voltage across pins U and V of table II shall be a minimum and the voltage across J and U or V of table II shall be less than the voltage applied to pins A and D of table III. The voltage across pins U and W and across V and W of table II shall be measured as the reference assembly is rotated about the roll gimbal in right roll. The voltage across pins U and W shall increase before decreasing and the voltage across pins V and W shall decrease before increasing.

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TABLE II. Displacement Gyroscope Assembly Output Connector (J361)

Bendix P/N PT02H-18-32P, or equal, to mate with Bendix P/N PT06P-18-32S, or equal	
Pin	Function
A	Spare
B	Spare
C	Spare
D	Roll switch - 90 - 270
E H Rotor	Azimuth CX, phase C, TOP
F C Rotor	Azimuth CX, BOTTOM
G H Rotor	115V ac excit - F to G in phase with (X) to (Z) at zero displacement
H C Rotor	Roll CX - DG
J H Rotor	115V ac excit - H to J in phase with (Z) to (X) at zero roll
K C Rotor	Pitch CX - EXT
L H Rotor	115V ac excit - K to L in phase with (Z) to (X) at zero pitch
M X Stator	Azimuth CX - BOTTOM
N Y Stator	
P Z Stator	
R X Stator	Azimuth CX - TOP
S Y Stator	
T Z Stator	
U X Stator	Roll CX - DG
V Y Stator	
W Z Stator	
X X Stator	Roll CX - VG
Y Y Stator	(Internally excited rotor - neutral to phase C in phase with (Z) to (X) at zero roll)
Z Z Stator	Pitch CX - EXT
a X Stator	
b Y Stator	
c Z Stator	Pitch CX - INT
d X Stator	(Internally excited rotor - netural to phase C in phase with (Z) to (X) at zero pitch)
e Y Stator	Inner roll CX
f Z Stator	(Internally excited rotor - neutral to phase C in phase with (Z) to (X) at zero displacement)
g X Stator	
h Y Stator	
j Z Stator	

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TABLE III. Displacement Gyroscope Assembly Input Connector (J362)

Bendix P/N PT02H-20-41P, or equal, to mate with Bendix P/N PT06P-20-41S, or equal	
Pin	Function
A (Neutral)	Input power - 3-phase 400 Hz, 4 wire
B (Phase A)	200V line-to-line/115V line-to-neutral
C (Phase B)	Phase A, phase B, and phase C phase Rotation
D (Phase C)	
E	Spare
F	Leveling signal
G	Leveling excitation
H	Leveling torquer control phase
J	Leveling torquer fixed phase
K	Pitch torquer A
L	Pitch torquer B
M	Pitch switch A
N	Pitch switch B
P	Pitch switch common
R	Pitch capacitor
S	Power failure
T	Pitch segment switch
U	Slaving torquer fixed phase
V	Slaving torquer cont phase
W	Pitch servo CT (H)
X	Reserved
Y	Reserved
Z	Pitch generator output
a	Roll switch B
b	Roll switch A
c	Roll torquer excitation
d	Roll torquer B
e	Roll torquer A
f	Roll generator output (hot)
g	Roll generator output (cold)
h	Pitch generator output (hot)
i	Leveling signal (A)
j	Leveling signal (B)
k	Reserved
m	Pitch M-G control phase
n	Pitch M-G fixed phase
p	Pitch generator excitation
q	Roll M-G control phase
r	Roll M-G fixed phase
s	Reserved
t	Roll generator excitation

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3.8.1.5.1.3 Inner roll synchro. Pin j of table II shall be connected to pin A of table III. The voltage g and h of table II and the voltage across pin D of table III and pin g or h of table II shall be measured with the gimbals at zero displacement. The voltage across pins g and h shall be a minimum and the voltage across pin D of table III and pin g or h of table II shall be less than the voltage applied to pins A and D of table III. The voltage across pins g and j and across h and j of table II shall be measured as the first gimbal (inner roll) is held and the reference assembly with the outer roll gimbal is rotated in right roll. The voltage across pins g and j of table II shall increase before decreasing and the voltage across pins h and j shall decrease before increasing.

3.8.1.5.1.4 Internally excited pitch synchro. Pin f of table II shall be connected to pin A of table III. The voltage across d and e of table II and the voltage across pin D of table III and pin d or e of table II shall be measured with the gimbals at zero displacement. The voltage across pins d and e shall be the minimum and the voltage across pin D of table III and pin d or e of table II shall be less than the voltage applied to pins A and D of table III. The voltage across pins d and f and across e and f of table II shall be measured as the second gimbal (pitch) is held and the reference assembly with the outer roll gimbal is rotated in climb. The voltage across pins d and f of table II shall increase before decreasing and the voltage across pins e and f shall decrease before increasing.

3.8.1.5.1.5 Internally excited outer roll synchro. Pin Z of table II shall be connected to pin A of table III. The voltage across X and Y of table II and the voltage across pin D of table III and pin X or Y of table II shall be measured with the gimbals at zero displacement. The voltage across pins X and Y shall be the minimum and the voltage across pin D of table III and pin X or Y of table II shall be less than the voltage applied to pins A and D of table III. The voltage across pins X and Z and across Y and Z of table II shall be measured as the third gimbal (outer roll) is held and the reference assembly is rotated in right roll. The voltage across pins X and Z of table II shall increase before decreasing and the voltage across pins Y and Z shall decrease before increasing.

3.8.1.6 Drift. Unless otherwise specified, the vertical gyro assembly shall be so designed that the maximum free drift, noted after compensation for the earth's rotation, shall be not greater than 0.25° per minute when operated over the entire temperature range specified herein, with the gyro spin axis tilted either 20° or vertical.

3.8.1.7 Power-off freedom. When power is removed momentarily, the vertical gyro shall still have freedom about its first and second gimbal axes and shall not be displaced from the vertical during bank angles of ±80° or pitch angles through ±80° during the power-off condition.

3.8.1.8 Roll segment switch. A roll segment switch shall be incorporated on the third gimbal axis. The switch shall close terminal D of table II to neutral from +90° through 180° to 270° of roll where zero roll is the initial zero position determined by 3.8.1.5.1.5. The circuit shall be open from the 270° roll position through zero to the +90° roll position.

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3.8.2 Directional gyro. The directional gyro shall have sufficient angular momentum and be of such design as to provide the performance specified herein when operated throughout the environmental ranges specified.

3.8.2.1 Gimbal system. The directional gyro shall be suspended in a gimbal system of which the outer gimbal is the third (servoed) gimbal of the vertical gyro. The directional gyro shall have three other gimbals suspended from the outer gimbal.

3.8.2.1.1 Inner directional gyro gimbal. The first (inner) directional gyro gimbal (gyro wheel housing) shall be stabilized by the inertia of the gyro. Its pivot axis shall be the first directional gyro axis of freedom, shall be perpendicular to the spin axis of the gyro, and normally shall lie in the horizontal plane. The slaving torquer shall act about this axis. The gimbal shall have a minimum of ±80° freedom about its axis.

3.8.2.1.2 Directional gyro azimuth gimbal. The second directional gyro gimbal shall be stabilized by the inertia of the gyro. Its pivot axis shall be aligned with the vertical. The azimuth output synchros and the leveling torquer shall act about this axis. The gimbal shall be capable of continuous rotation about its axis.

3.8.2.1.3 Directional gyro pitch gimbal. The third directional gyro (pitch) gimbal shall be servo driven so that it follows the vertical gyro second (pitch) gimbal at a maximum rate of not less than 200° per second. Its pivot axis shall be the third directional gyro axis of freedom and shall be parallel to the vertical gyro second axis (pitch) of freedom. The gimbal shall be capable of continuous rotation about its axis.

3.8.2.2 Directional gyro leveling. The inner gimbal shall be slaved by means of the leveling torquer controlled by a suitable non-pendulous sensor so that the spin axis of the directional gyro is perpendicular to the azimuth axis. This leveling rate shall have an initial rate that will level the gyro within 1-1/2 minutes after power is applied. It shall have a normal leveling rate of 2° to 6° per minute.

3.8.2.3 Directional gyro slaving. The directional gyro shall be capable of being precessed about its azimuth axis to an external azimuth signal by means of the azimuth torquer. The normal slaving rate shall be 1° to 2° per minute.

3.8.2.4 Azimuth transmitting synchros. Two synchros shall be attached to the azimuth axis of the gyro to transmit signals proportional at a 1-to-1 ratio to the position of the gyro. The synchros shall not exceed a maximum transmission error of 8 minutes of arc. The synchros shall be of the 3-phase wye-connected stator and salient-pole single-phase rotor types. The synchros shall have a no-load line-to-line output of 11.8 ±0.25V and a phase shift of 4.0° ±1.5°. When loaded with 8 CT's, each having a 10,000-ohm load, the line-to-line output voltage shall be not less than 10.6V and the phase shift shall be not more than 10°.

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3.8.2.4.1 Azimuth transmitter. The azimuth transmitter synchro shall be capable of operating up to eight high-impedance control transformers of the Eclipse-Pioneer AY500-5 type, or equal, when the control transformers are loaded with 10,000-ohm loads. The stator leads of the synchro shall be brought out to the pins of the signal connector in accordance with table II and shall be separate from other circuitry. The synchro rotor shall be internally connected to neutral and shall be excited by the voltage applied to pin E as shown in table II.

3.8.2.4.2 Autopilot transmitter. The autopilot transmitter shall operate from a rotor excitation of 115V 400 Hz and the maximum stator output voltage, line-to-line, shall be 11.8V. The transmitter shall have an electrical accuracy of ±8 minutes or less. The rotor and stator leads shall be brought out to separate pins of the electrical connector as shown in table II, and shall be isolated from other circuitry.

3.8.2.4.3 Synchro zeroing and sensing. Pin T of table II shall be connected to pin A of table III. The voltage across R and S of table II, and the voltage across D of table III and R and S of table II shall be measured with the gimbals at zero displacement, the spin axis of the directional gyro aligned with the fore-aft axis of the displacement gyroscope assembly, and with the directional gyro spinning clockwise (cw) when viewed from the aft end. The voltage across pins R and S of table II shall be at a minimum and the voltage across pin D of table III and pin R or S of table II shall be greater than the voltage applied to pins A and D of table III. The voltage across pins R and T and across S and T of table II shall be measured as the directional gyro second (azimuth) gimbal is held and the reference assembly is rotated about its axis in right turn. The voltage across pins R and T of table II shall increase before decreasing and the voltage across pins S and T of table II shall decrease before increasing. During this procedure, pin E of table II shall be connected to pin D of table III.

3.8.2.4.3.1 Externally excited autopilot transmitter. Pins P and F of table II shall be connected to power lead pin A of table III, and pin G of table II shall be connected to power lead pin D of table III. The voltage across pins M and N of table II and the voltage across pins G and M or N of table II shall be measured with the gimbals at zero displacement, the spin axis of the directional gyro aligned with the fore-aft axis of the reference assembly, and with the directional gyro spinning cw when viewed from the aft end. The voltage across pins M and N of table II shall be a minimum and the voltage across pin D of table III and pin M or N of table II shall be greater than the voltage applied to pins A and D of table III. The voltage across pins M and P and across N and P of table II shall be measured as the directional gyro second (azimuth) gimbal is held and the reference assembly is rotated about its azimuth axis in right turn. The voltage across pins M and P of table II shall increase before decreasing and the voltage across pins N and P of table II shall decrease before increasing.

3.8.2.5 Balance. The directional gyro shall be so balanced that the apparent drift due to earth's rotation will be equal to the product of the sine of the latitude of operation and 15° per hour in a counterclockwise (ccw) direction in the Northern Hemisphere and in a cw direction in the Southern Hemisphere.

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3.8.2.6 Maximum drift. The maximum drift of the directional gyro shall not exceed ±1.0° per hour from the computed apparent drift for the latitude of operation at room temperature.

3.8.2.7 Random drift. With normal distribution, the random drift of the directional gyro shall not exceed ±0.25° per hour - one sigma.

3.8.2.8 Mean drift. The mean drift of the directional gyro shall not exceed ±1.5° per hour for the computed apparent drift for the latitude of operation over a temperature range of -54° to +71°C.

3.8.3 Electrical connectors. The electrical connectors shall be in accordance with tables II, III, and IV and figure 1.

TABLE IV. Displacement Gyroscope Assembly Calibration Connector (J363)

Bendix P/N PT02H-12-10P, or equal, to mate with Bendix P/N PT06P-12-10S, or equal	
Pin	Function
A	Pitch torquer CT
B	Pitch torquer
C	Pitch torquer
D	Roll torquer CT
E	Roll torquer
F	Roll torquer
G	Spare
H	Spare
J	Spare
K	Spare

3.8.3.1 Calibration connector (J363). The calibration connector shall be supplied with a metal cover that will remain in place during service use. Means shall be provided to mechanically secure the cover to the displacement gyroscope case to prevent loss of the cover during service usage.

3.8.4 Weight. The weight of the displacement gyroscope assembly shall not exceed 26-1/2 pounds including mounting base.

3.8.5 Dimensions. The dimensions of the displacement gyroscope assembly shall be in accordance with figure 1.

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3.8.6 Mounting. The mounting of the displacement gyroscope assembly shall be in accordance with figure 1 and such that the displacement reference will not deflect by more than ±0.25° in pitch and roll or 0.02° in azimuth as a result of accelerations or any vibration between 0 and 18 Hz with a minimum amplitude of 0.020 inch. The torsional resonant period shall be greater than 18 Hz.

3.8.6.1 Means shall be provided for quick release of the displacement gyroscope assembly, including mount from the aircraft.

3.8.6.2 Means shall be provided for precision adjustment of the displacement gyroscope assembly in relation to its mount.

3.8.7 Elapsed time indicator. The displacement gyroscope assembly shall contain an MS24534-3D elapsed time indicator with a range of 3,000 hours which shall be mounted as shown on figure 1.

3.9 Amplifier-power supply. The amplifier-power supply shall fulfill the necessary servo and power requirements for proper normal gyroscope operation. It shall contain the amplifiers for the servoed gimbals and related equipment and the starting and erection controls. Its connectors and pin functions shall be in accordance with tables I and V.

3.9.1 Amplifier channels. The amplifier channels shall be interchangeable, plug-in type where practicable and where reliability is not degraded by such design requirements. The amplifiers shall not incorporate electronic tubes. Parallel circuits shall be used where practicable so that partial failure will not result in total loss of operation.

3.9.2 Starting and erection control. The starting and erection control shall be designed to perform the functions of starting sequence control, fast erection control, power failure warning, roll erection cutout, and pitch erection cutout.

3.9.2.1 Starting sequence control. The starting sequence control shall perform the function of properly aligning the gimbals of the displacement gyroscope assembly and erecting and aligning the gyros as required in 3.8.1.4.1, 3.8.1.4.2, and 3.8.2.2 when the servoed gimbals are in any initial position and when the system power has been disconnected long enough to cause recycling even though the gyros may still be coasting. Proper alignment shall also occur when the servoed shafts and gimbals of the gyroscope assembly are rotated to any new position after power is disconnected.

3.9.2.2 Power-failure warning. A 3-phase power failure warning circuit shall be incorporated in the control assembly. This circuit shall operate a set of normally open contacts. Leads from the set of contacts shall be brought out of the case to pins K and L of table I separate from other circuitry. When proper power is applied to the control, the contacts shall close the circuit between pins K and L after the initial erection cycle is completed. If at any time any phase of the 3-phase power to the control is broken, the contacts shall open breaking the circuit between pins K and L.

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TABLE V. Amplifier-Power Supply Output Connector (J201)

Bendix P/N PT00P-20-41PY, or equal, to mate with Bendix P/N PT06P-20-41SY, or equal	
Pin	Function
A (Neutral)	115/200V AC, 400 Hz
B (Phase A)	3-phase 4-wire WYE
C (Phase B)	Phase rotation A-B-C
D (Phase C)	
E	Spare
F	Leveling input
G	Leveling excitation
H	Leveling control phase
J	Leveling fixed phase
K	Disable alarm relay
L	Spare
M	Spare
N	Spare
P	Pitch erection voltage
R	Pitch erection voltage
S	Test - unfiltered dc
T	Test - filtered dc
U	Pitch alarm
V	Spare
W	Pitch servo CT (H)
X	Roll M-G fixed phase (w/o third gimbal controller)
Y	Spare
Z	Power failure
a	Roll servo input
b	Start cycle - 60 sec
c	Roll erection voltage
d	Roll servo input (w/o third gimbal controller)
e	Roll servo input
f	Roll servo input
g	Start cycle - 60 sec
h	Pitch generator output (signal ground)
i	Leveling input
j	Leveling input, common
k	Spare
m	Pitch M-G control phase
n	Pitch M-G fixed phase
p	Pitch generator excitation
q	Roll M-G control phase
r	Roll M-G fixed phase (w/third gimbal controller)
s	Spare
t	Roll generator excitation

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3.9.3 Case. The case for the amplifier-power supply shall conform to figure 3.

3.9.4 Mounting and mounting base. The mounting of the amplifier-power supply shall be such that the case can be connected directly to the mounting base or such that a vibration isolator can be inserted between the case and mounting base. The mounting base shall conform to figure 4 and shall be optional (see 6.2).

3.9.5 Weight. The weight of the amplifier-power supply shall not exceed 6.5 pounds without the mounting base and shall not exceed 7.75 pounds with mounting base.

3.9.6 Elapsed time indicator. The amplifier-power supply shall contain an MS17322-6A elapsed time indicator with a range of 10,000 hours.

3.10 Compass adapter. The compass adapter unit shall contain the azimuth servo with amplifier, azimuth slaving amplifier, fast synchronization and latitude correction circuitry, and synchros as required.

3.10.1 Amplifier channels. The amplifier channels shall be the plug-in type and interchangeable, where practicable. The channels shall be rugged and reliable and shall be suitable for a minimum of 1,500 hours of operation. The amplifier shall not incorporate electronic tubes.

3.10.2 Fast synchronization and heading set drive. An electro-mechanical mechanism shall be provided to position the stator of the differentially driven azimuth servo control transformer and to position the rotor of the attitude indicator differential synchro the same amount. The mechanism shall cause synchronization of the azimuth servo channel when operating in the SLAVED mode and shall position the azimuth servo to any desired reference to set the heading when the heading system is operating in the DG mode. The mechanism shall have a drive rate continuously variable in either direction 3° to 50° per second as selected by the manual synchronizer and heading set control specified in 3.11.2. The mechanism shall be suitably damped to permit accurate synchronization manually or automatically and accurate selection of the desired heading reference.

3.10.2.1 Attitude indicator differential synchro. The attitude indicator differential synchro shall have all six leads brought out of the case separately to pins L, M, N, P, R, S of table VI. When its rotor leads L, M, N are properly connected to the stator leads of the gyro azimuth transmitter, the output stator signals shall be synchronized with the azimuth servo data transmitter except for: (1) transient servo error of the azimuth servo and (2) synchro nulling when the azimuth servo null is reversed by a DCK-10/A24G-1 third gimbal controller. The stator leads shall function properly with a high-impedance control transformer synchro as specified in 3.8.1.5.

3.10.2.1.1 Differential synchro zeroing. The differential synchro shall be zeroed so that when the displacement gyroscope gimbals and the azimuth servo data transmitter are at zero displacement, the voltage across pins P and R of table VI shall be a minimum and the power voltage from neutral to phase C shall be in phase with the voltage from pins P to S of table VI.

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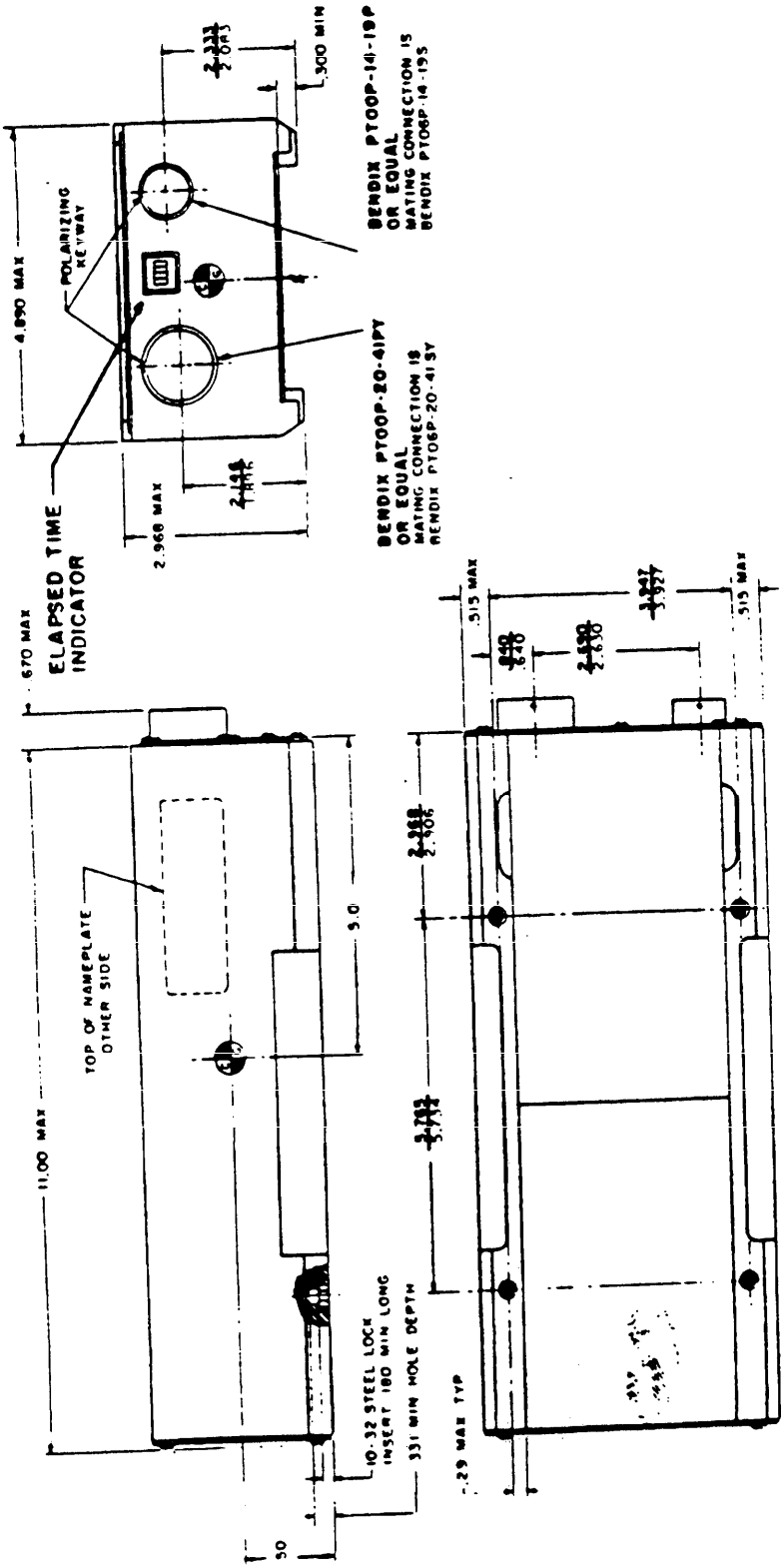


FIGURE 3. Power Supply Amplifier

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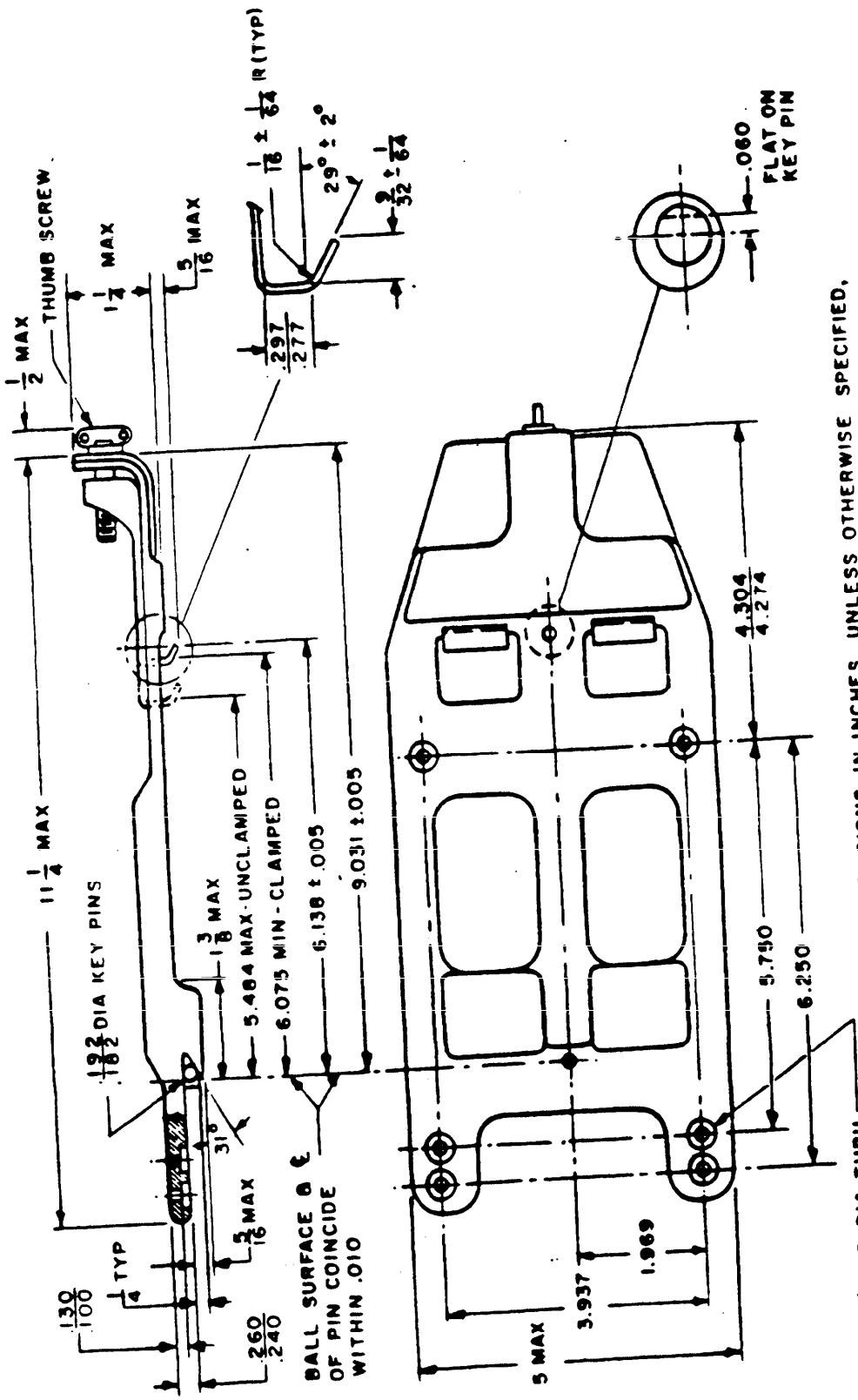


FIGURE 4. Mounting Base

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TABLE VI. Compass Adapter Output Connector (J902)

Bendix P/N PT07H-20-41 PW, or equal, to mate with Bendix P/N PT06P-20-41 SW, or equal	
Pin	Function
A	Tachometer
B	Tachometer
C	Spare
D	Spare
E	Shielding ground
F	Spare
G	Spare
H	Spare
J	26V synchro excitation
K	Terminal J to K in phase with neutral to phase C
L X Rotor	
M Y Rotor	
N Z Rotor	Differential transmitter
P X Stator	B208 - (3.10.2.1)
R Y Stator	
S Z Stator	
T H Rotor	
U C Rotor	Output data transmitter
V X Stator	B205 - (3.10.8.1)
W Y Stator	28V excitation
X Z Stator	
Y H Rotor	
Z C Rotor	Output data transmitter
a X Stator	B202 - (3.10.8.1)
b Y Stator	28V excitation
c Z Stator	
d H Rotor	
e C Rotor	Output data transmitter
f X Stator	B203 - (3.10.8.1)
g Y Stator	26V excitation
h Z Stator	
i H Rotor	
j C Rotor	Output data transmitter
k X Stator	B204 - (3.10.8.1)
m Y Stator	26V excitation
n Z Stator	
p	23V ac flux valve excitation
q	p to q in phase with phase A to neutral
r X Stator	
s Y Stator	Flux valve control
t Z Stator	transformer

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3.10.2.2 Automatic synchronization. The azimuth servo shall be automatically synchronized with the DT-173/AJN detector, or electrical equivalent, upon initial application of power to the system when the control panel function selector switch is in the SLAVED position and when the heading system is operating and is switched from the DG mode of operation to the SLAVED mode. Automatic synchronization shall be accomplished to within ±2° at a rate greater than 50° per second.

3.10.2.2.1 Push-to-sync circuit. The design of the circuit shall be such that when the PUSH-TO-SYNC button (see 3.11.2.1) is pressed and held with the heading system in the SLAVED mode of operation, the system will synchronize at the rates and accuracies specified for automatic synchronization.

3.10.2.3 Automatic autopilot decoupling. The continuity between pins F and G of table VII and figure 5 shall be transferred to between G and H incident to and during the following conditions only. The design shall be such that the transfer must be made before the automatic fast synchronizations can be actuated.

a. Synchronization
b. Slewing
c. Compass mode operation

3.10.3 Latitude correction. A signal controlled as specified in 3.11.3 shall correct the output data transmitters for the apparent drift of the gyro due to the earth's rotation. The correction shall be at a rate equal to the product of 15° per hour and the sine of the indicated latitude within a tolerance of ±1.0° per hour. The correction shall be cw in the Northern Hemisphere and ccw in the Southern Hemisphere. The design shall be such that a proper external precision voltage can be used to provide the correction signal as a function of the exact latitude.

3.10.4 Azimuth servo. The azimuth servo channel shall contain a servo system with a follow-up rate of at least 30 rpm to maintain synchronization with the gyro azimuth signal throughout all maneuvers. The operation shall be smooth with no lag in excess of 0.2° per rpm up to 10 rpm. Adequate damping shall be provided so that no oscillation, hunting, or jumping of a magnitude or frequency that will affect the operation of a high-performance automatic flight control system shall occur. The undamped natural frequency shall be greater than 6 Hz. The control transformer of the azimuth servo will receive its signals from the directional gyro azimuth transmitter. The azimuth servo shall position the rotors of four data transmitter synchros and the slaving channel control transformer.

3.10.4.1 Azimuth servo control transformers. The azimuth servo control transformer shall be differentially mounted to provide the operation specified in 3.10.2. Its stator and rotor leads shall be brought out of the case through pins p, q, r, s, and t of table VII separate from other circuitry and each other. The alignment of components and circuitry of the azimuth servo shall be that required to permit control transformer pins r, s, and t of table VII to be connected to pins R, S, and T of table II, respectively, and to permit the rotor leads, pins p and q of table VII to be connected to pins m and n

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TABLE VII. Compass Adapter Input Connector (J901)

Bendix P/N PT07H-20-41 PX, or equal, to mate with Bendix P/N PT06P-20-41 SX, or equal	
Pin	Function
A (Neutral)	115/200V AC, 400 Hz
B (Phase A)	3-phase 4-wire WYE
C (Phase B)	Phase rotation A-B-C
D (Phase C)	
E	External switch at slaving cutout - N. O.
F	Autopilot decouple - N. C.
G	Autopilot decouple - ARM.
H	Autopilot decouple - N. O.
J	Autopilot decouple - compass mode
K	Spare
L	Compass mode relay
M	Neutral
N	External switch at slaving cutout - N. C.
P	External switch at slaving cutout - ARM
R	Slaving common - neutral
S	Reserved
T	Reserved
U	Lat Corr excitation
V	Lat Corr excitation
W	Push-to-turn relay ground
X	Push-to-turn relay ground
Y	Set. hdg Sig input
Z	Set. hdg excitation
a	Set. hdg excitation
b	Manual sync - PUSH TO SYNC
c	Slave mode relay ground
d	Reserved
e	Reserved
f	Sync meter
g	Sync meter
h	Reserved
i	Reserved
j	Lat corr sig
k	Slaving cutout
m	Azimuth servo input
n	Azimuth servo input - (i.g. ground)
p Rotor	Gyro CT output
q Rotor	Gyro CT output
r X Stator	
s Y Stator	Gyro control transformer
t Z Stator	B206 - (see 3.10.4.1)

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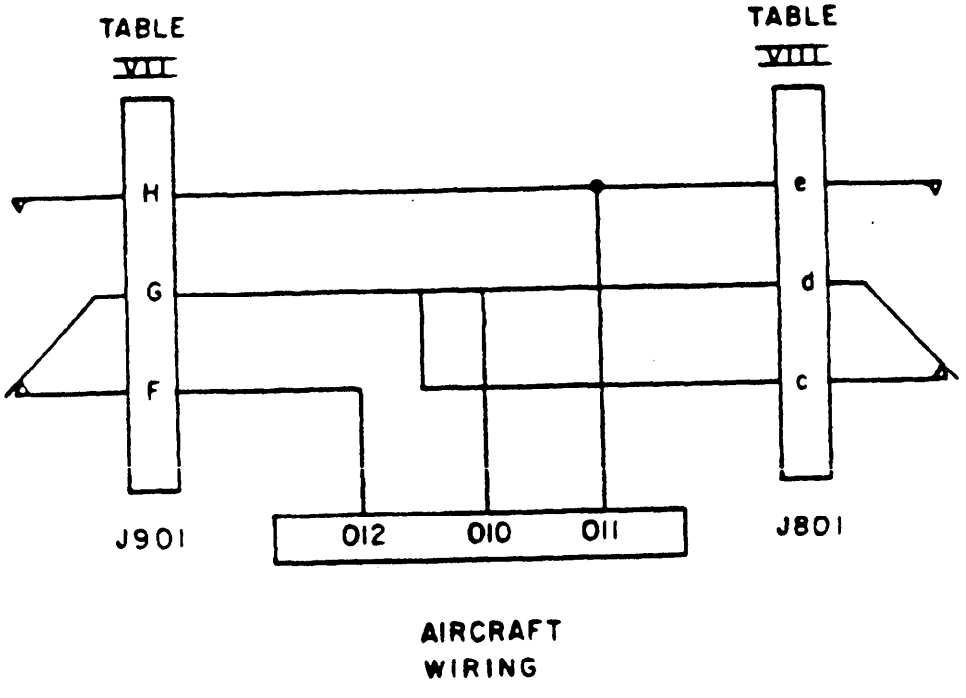


FIGURE 5. Interconnect For Autopilot Decoupling

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of table VII, respectively, for proper operation of the servos. The design shall be such that when these rotor leads are externally reversed, the azimuth servo shall rotate 180° to the reciprocal of its position before the reversing.

3.10.4.2 Slaving channel control transformer. The slaving channel control transformer shall be attached to the azimuth servo and so aligned that when the control assembly is connected properly and operating, the direction gyro azimuth transmitter is at its reference position as specified in 3.8.2.4.3 and when the DT-173/AJN detector is connected to the control transformer and is aligned with magnetic north, the azimuth servo data transmitters will be at zero displacement and the system will be synchronized.

3.10.5 Slaving loop. The slaving loop shall cause the output data shaft to align to the DT-173/AJN detector signal at a nominal rate of 1° to 2° per minute

3.10.6 Compass mode operation. Compass mode operation shall cause the output data shaft to align to the DT-173/AJN detector signal. The compass mode shall be initiated by connecting pins c, J, and L to pin A of table VII. Compass mode operation shall not be affected by the input to the azimuth servo control transformer (pins r, s, and t of table VII).

3.10.7 Slaving cutout. Application of 115V ac 400 Hz to pin k of table VII shall cause the operation specified in 3.10.5 to be interrupted. The operation specified in either 3.10.3 or 3.10.6 shall not be interrupted. The application of 115V ac may be accomplished by connecting pin k of table VII to pin P of table I, with or without pins M and N of table I also being connected.

3.10.8 Data transmitters. Four data transmitters whose rotors are positioned by the azimuth servo shall be provided for the electrical transmission of heading reference to auxiliary equipment. All connections to rotors and stators shall be brought out to the terminals as shown in table VI and shall be separate from each other and from other circuitry. The zeroing and sensing of each synchro shall be in accordance with the procedure in 3.10.8.2 and the stator and rotor leads shall correspond to table VI.

3.10.8.1 Precise data transmitters. Four transmitter synchros shall be provided for operating up to eight precise receiver synchros of the high-impedance control-transformer type, such as the Eclipse-Pioneer AY-500-5 or equal, when the rotors of the control transformers are loaded with 10,000 ohms minimum. The transmitter synchros shall have a room temperature no-load line-to-line output of 11.8 ±0.5V, a phase shift of 9.5° ±3.0°, and an accuracy of ±7 minutes of arc or less. When loaded with four control transformers each having a 10,000-ohm load, the room temperature line-to-line output voltage shall be not less than 10.9V, the phase shift shall be not more than 15.2°, and the accuracy shall be ±8 minutes of arc or less. When loaded with eight control transformers each having a 10,000-ohm load, the room temperature line-to-line output voltage shall be not less than 10.2V, the phase shift shall be not more than 19.0°, and the accuracy shall be ±11 minutes of arc or less .

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3.10.8.2 Data transmitter zeroing and sensing. The data transmitters shall each be zeroed and their sensing established in accordance with the following procedure. Their leads shall be brought out of the case separately and identified in accordance with table VI.

3.10.8.2.1 The rotor leads of the data transmitter synchro shall be identified as H and C and the stator leads shall be identified as X, Y, and Z. The gyroscope gimbals shall be at zero displacement. The flux-valve type signal which accurately simulates magnetic north shall be imposed on the slaving channel control transformer and the system shall be synchronized. The data transmitter synchro shall be excited from C to H with the proper voltage in phase with the power voltage from neutral to phase C. The voltage from X to Y shall be at a minimum and with rotor lead C connected to stator lead Z, the voltage from H to X shall be greater than the excitation voltage. The voltage between X and Z shall increase before decreasing, and between Y and Z shall decrease before increasing for cw rotation of the displacement gyroscope about the azimuth axis from the zero displacement position (increasing azimuth function). For opposite rotation (decreasing azimuth function), the voltage changes shall be opposite.

3.10.9 Excitation voltage output. The compass adapter shall provide a 26V a-c output from pins J and K of table VI which shall be in phase with the voltage applied from pins A to D of table VII and a 23V a-c output from pins p to q of table VI. The voltage from p to q of table VI shall be in phase with the voltage applied from pins B to A of table VII. The former voltage shall be suitable for the excitation of the data transmitter synchros and the associated repeater indicators. The latter voltage shall be for excitation of the flux valve.

3.10.10 Case. The compass adapter case shall be in accordance with figure 6.

3.10.11 Weight. The weight of the compass adapter without mounting base shall not exceed 7-1/4 pounds and with mounting base shall not exceed 8-1/2 pounds.

3.10.12 Mounting and mounting base. The mounting of the unit shall be such that the case can either be connected directly to the mounting base or such as to permit a vibration isolator to be inserted between the case and mounting base. The mounting base shown on figure 4 shall be optional (see 6.2).

3.10.13 Elapsed time indicator. The compass adapter shall contain an MS17322-6A elapsed time indicator with a range of 10,000 hours.

3.10.14 Deviation compensation. Provisions shall be made for electrically adding up to ±3.5° deviation between the flux valve and the transmitters at 24 points from 0° to 345° in 15° increments. Each point shall be continuously variable for 0° ±3.5° deviation and shall be independent of the other 23 points.

3.10.15 Compass mode. Provisions shall be made for slaving the four data transmitters specified in 3.10.8.1 directly to the magnetic azimuth detector independent of azimuth gyro damping action.

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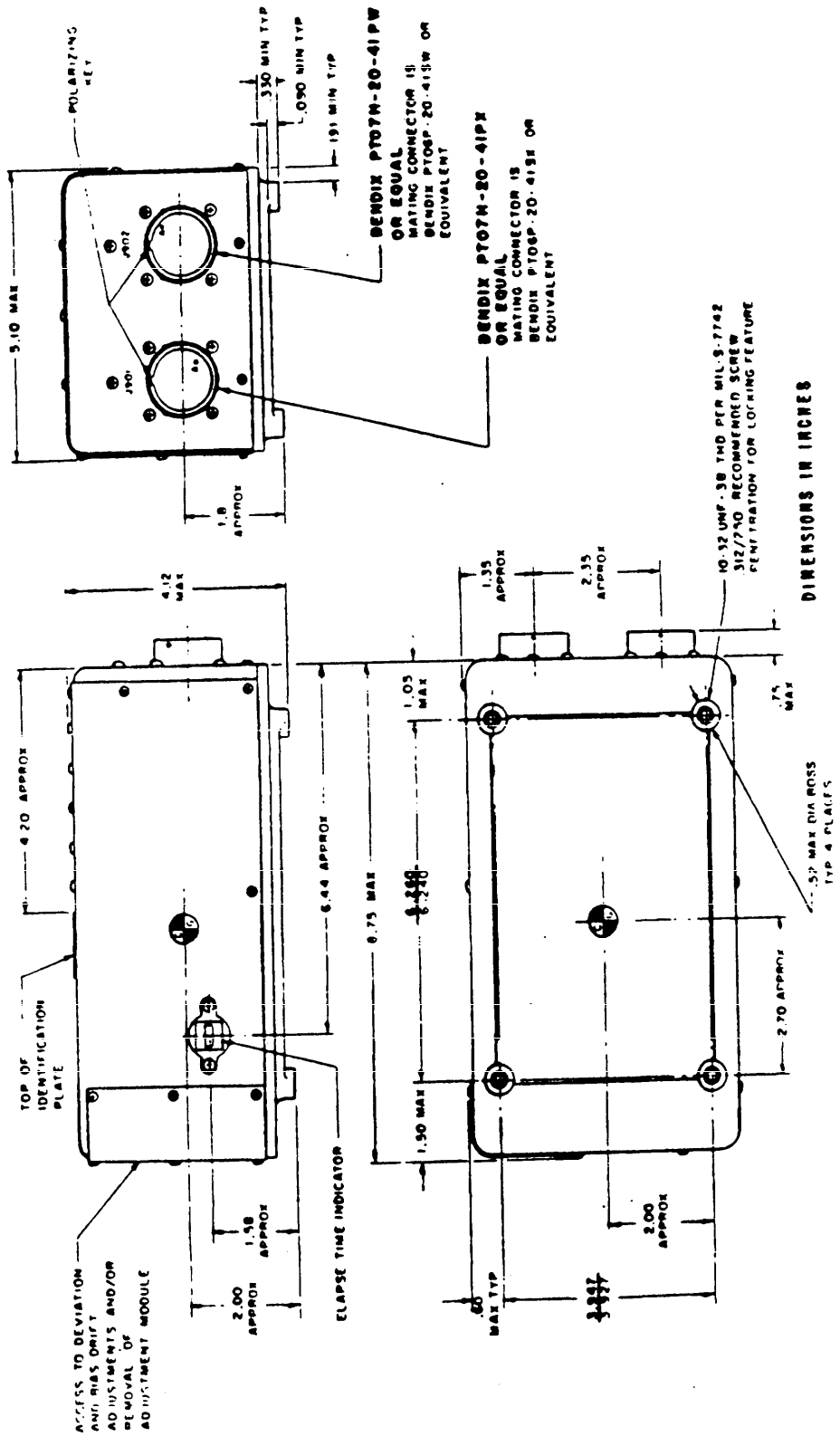


FIGURE 6. Compass Adapter

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3.11 Compass controller. A compass controller designed in accordance with MIL-C-6781, MIL-P-7788, and MIL-F-25173 shall provide the controls and indications necessary to operate the heading system. The compass controller shall conform to figure 7. This unit may be gasket sealed.

3.11.1 Function selector switch. A switch shall be provided to select the compass system mode of operation. When the switch is in the SLAVED position, the system shall operate as a gyro stabilized magnetic compass and synchronize with the DT-173/AJN detector, or electrical equivalent, and pin K of table VIII shall be open. When the switch is in the DG position, the system shall operate as a latitude-corrected directional gyro, and pin K of table VIII shall be connected internally to the neutral power lead.

TABLE VIII. Compass Controller Connector (J801)

Bendix P/N PT02E-18-32P, or equal, to mate with Bendix P/N PT06P-18-32S, or equal	
Pin	Function
A	Neutral
B-J	Spare
K	Doppler cutout
L	Slaving common
M	Latitude common
N	Lat Comp supply cont phase
P	Lat Comp supply cont phase
R	Sync motor automatic
S	Sync motor automatic
T	Sync motor supply
U	Sync motor manual
V	Sync motor manual
W	Manual sync
X	Auto sync time delay
Y	Slaving fixed phase
Z	Slaving fixed phase
a	Discriminator
b	Discriminator
c	Manual slew autopilot decouple
d	Manual slew autopilot decouple
e	Manual slew autopilot decouple
f	Lat compensation fixed phase
g	Lat compensation fixed phase
h	Slaving control phase
j	Dimmer control

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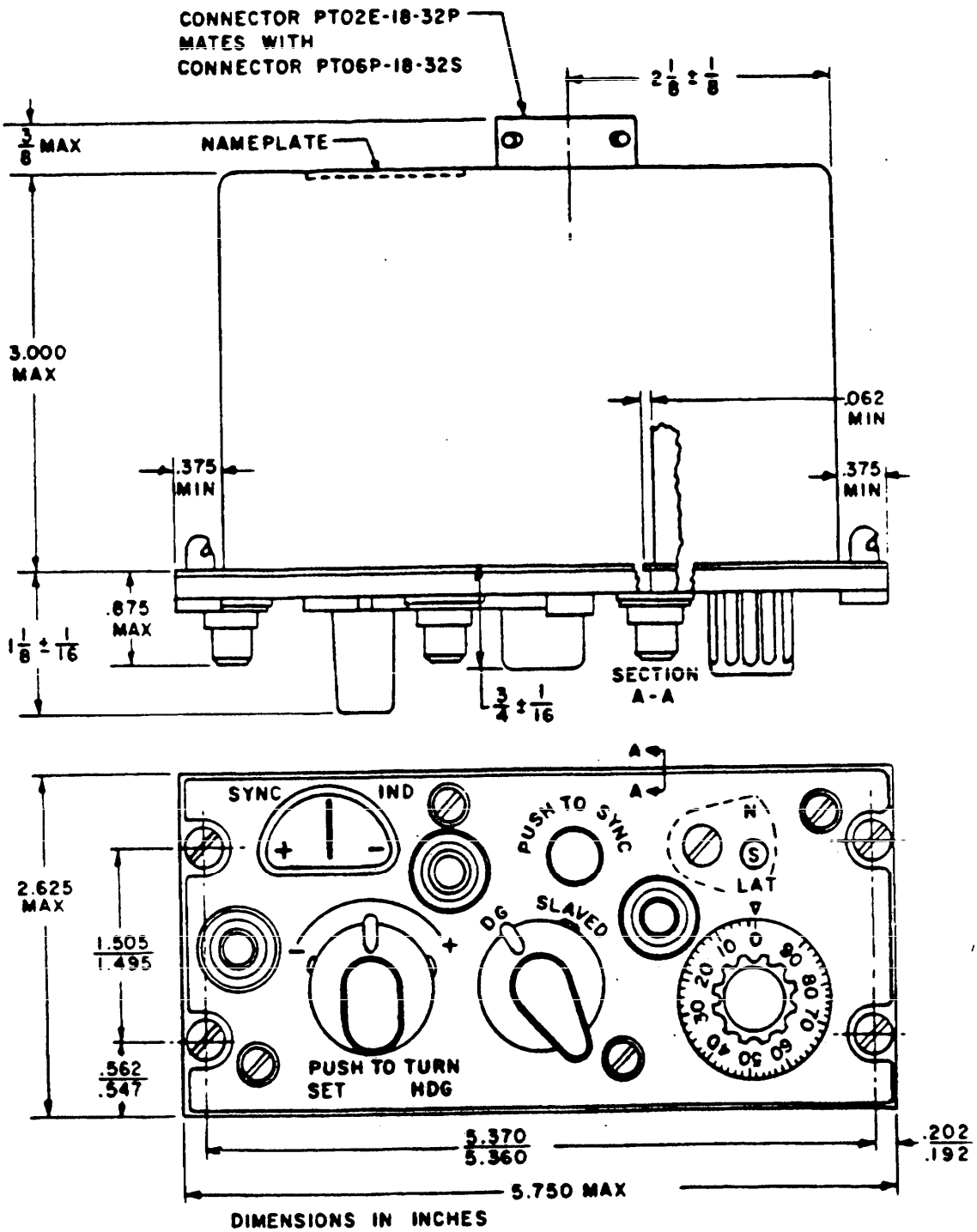


FIGURE 7. Controller, Compass

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3.11.2 Synchronizer and heading set. A manual control shall be provided to affect the fast synchronization and heading set as specified in 3.10.2. An accurate indicator, properly connected to pins V, W, X of table VI to indicate increasing and decreasing azimuth function, shall show increasing function when the synchronization control is rotated in a cw direction during either SLAVED or DG mode of operation. The synchronization control markings shall correspond to the annunciator markings and shall indicate in which direction the control is to be turned to synchronize the system when in the slaved mode of operation. The control shall be of the push-to-turn, spring return-to-center type.

3.11.2.1 PUSH-TO-SYNC button. A push button switch shall be incorporated for actuating the PUSH-TO-SYNC circuit specified in 3.10.2.2.1.

3.11.2.2 Manual slew autopilot decoupling. During normal operation, continuity shall exist between pins c and d of table VIII and d to e shall be open. When the manual control is pressed in for manual synchronization or heading set, the continuity shall be transferred from between c and d to d and e.

3.11.3 Latitude correction controller. A manually set controller shall be provided to correct the system (when in the DG mode of operation) for the apparent drift of the gyro due to the earth's rotation as specified in 3.10.3. The selection of direction of the correction for the Northern or Southern Hemisphere shall be made by a hemisphere switch. The hemisphere switch shall be controlled from a shaft with a screwdriver slot for turning. The latitude dial shall be marked with latitude from 0° to 90° graduated in 2° increments, with major divisions and numerals every 10°.

3.11.4 Annunciator. An annunciator, other than signal lights, shall be provided to indicate synchronization to within ±1/4° between the compass system and the DT-173/AJN detector, or electrical equivalent, when in the magnetic mode of operation. When the system is not synchronized, the annunciator shall indicate in which direction the synchronizer must be turned to avoid an ambiguity. The annunciator lighting and markings shall afford ease of viewing under all ambient lighting conditions.

3.11.5 Markings and dimensions

3.11.5.1 Markings. All markings shall be so durable as to withstand usage encountered in service. Unless otherwise specified, the style and proportions of numerals and letters shall conform to MS33558.

3.11.5.2 Numerals. Each numeral shall be so distinctly marked that no doubt will exist as to which graduation the numeral applies.

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3.11.5.3 Dimensions. The dimensions of the markings shall be as follows:

	Height or Length Inch ±1/64	Width of Line or Graduation Inch ±0.005
Letters or words	1/8	0.015
Numerals	1/8	0.015
2° graduations	1/16	0.020
10° graduations	3/32	0.020
+ and -	1/8	0.020 to 0.040

3.11.5.4 Color. The background of the dial and all other markings not otherwise specified shall be finished in durable dull black.

3.11.6 Weight. The weight of the compass controller shall not exceed 1-1/4 pounds.

3.11.7 Mounting. The mounting of the compass controller shall be such as to enable the unit to meet the tests specified in 4.6.75 and 4.6.84.1.

3.12 Electronic components. Electronic components shall be in accordance with MIL-E-5400.

3.13 Electrical power. The power required by the control assembly shall be 4-wire 200/115V 3-phase 400-Hz (nominal) in accordance with MIL-STD-704 and as specified herein.

3.13.1 Electrical power variation. The control assembly shall operate properly for any a-c voltage from line-to-neutral between 103 and 127V, for any frequency between 380 and 420 Hz or any combination thereof.

3.13.2 Phase rotation. The phase rotation of the 400-cycle power required by the system shall be A-B-C, with neutral as ground.

3.13.3 Fusing. With the DCK-10/A24G-1 third gimbal controller connected, the control assembly shall operate properly with 1.5-amp fuses installed in each of the three power phases. Fuses shall be in accordance with MIL-F-15160.

3.14 Electrical connectors. Unless otherwise specified, electrical connectors shall be in accordance with MIL-C-5015.

3.14.1 Test terminals. Adequate test terminals shall be provided on each component for external testing of the item.

3.15 Cases. The cases shall be in accordance with the specified dimensions and shall provide a hermetically sealed inclosure when required for the internal mechanism. The design of the case shall be such that the internal mechanism may be removed

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from the case, replaced, and the case resealed without the use of special tools and fixtures unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the control assembly may be subjected and in such a manner that it will meet the test specified in 4.6.63.

3.15.1 Hermetic sealing. A hermetic seal is a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets. The following requirements shall apply except in those extreme cases where the filling medium or pressure, or both, materially affects the performance of the instrument. Exceptions shall be based upon sound engineering facts and shall be presented to the procuring activity for approval.

3.15.1.1 Filling medium. When a hermetically sealed case is used, the filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be either 100 percent helium, 95 percent helium and 5 percent carbon dioxide, or a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere or 1/2 atmosphere as required. Where practicable, the 100 percent helium filling medium shall be utilized. Other gases may be used when approved by the procuring activity.

3.16 Soldering. Soldering shall be accomplished in accordance with MIL-STD-454, requirement 5.

3.17 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.18 Finishes and protective coatings

3.18.1 Aluminum alloy parts. Aluminum parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.18.1.1 Dials, small holes, and case inserts need not be anodized.

3.18.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.18.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.18.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

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3.18.1.5 When abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.18.2 Cadmium plating. Cadmium plating shall be in accordance with QQ-P-416, of a type and class required to achieve the degree of protection required.

3.19 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.20 Workmanship. The control assembly, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machinescrew assemblies, and freedom of parts from burrs and sharp edges.

3.20.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the system, they shall be held or limited accordingly.

3.20.2 Screw assemblies. Assembly screw and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

3.20.3 Riveting. Riveting operations shall be performed carefully to insure that the rivets are tight and satisfactorily headed.

3.20.4 Gears. Gear assemblies shall be properly aligned and meshed and shall operate without interference, tight spots, loose spots or other irregularities. Where required for accuracy adjustment, gear assemblies shall be free from back lash.

3.20.5 Cleaning. The control assembly components shall be cleaned thoroughly of loose, spattered, or excess solder, metal chips, or other foreign material after assembly. Burrs and sharp edges as well as resin flash which might crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of tests. The inspection and testing of components of the control assemblies shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

4.3 Test conditions

4.3.1 Properly connected equipment. When a component or the control assembly is referred to as being properly connected, it shall be connected in accordance with the following, as applicable:

4.3.1.1 Displacement gyroscope assembly

4.3.1.1.1 A, B, and C phases of power as specified under 3.13 shall be connected to pins B, C, and D, respectively, with pin A neutral. Phase rotation shall be A, B, and C.

4.3.1.1.2 A voltage and frequency of 115V rms ±3 percent, 400 Hz ±5 percent, phase C to neutral, shall be applied across pins L and K of J361, respectively.

4.3.1.1.3 A voltage and frequency of 115V rms ±3 percent, 400 Hz ±5 percent, phase C to neutral, shall be applied across pins J and H of J361, respectively.

4.3.1.1.4 A voltage and frequency of 115V rms ±3 percent, 400 Hz ±5 percent, phase C to neutral, shall be applied across pins G and F of J361, respectively.

4.3.1.1.5 A voltage and frequency of 115V rms ±3 percent, 400 Hz ±5 percent, phase C to neutral, shall be applied across pins D and A of J362, respectively.

4.3.1.1.6 A voltage and frequency of 115V rms ±3 percent, 400 Hz ±5 percent, phase C to neutral, shall be applied across pins E of J361 and A of J362, respectively.

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4.3.1.1.7 Pitch servo amplifier and connections. The pitch servo amplifier and connections shall be as follows:

Gyro Reference		Lear Amplifier No. 131676-01, or equivalent
J362 W	to	Pin 4
J362 Z	to	Pin 5
J362 m	to	Pin 10
J362 A	to	Pins 3 and 8
		Pin 6 to Pin 2
J362 D	to	Pin 9

Pins d, e, and f of J361 shall be connected to pins k, Y, and s respectively, of J362. A voltage and frequency of 32V rms ±3 percent +90°, 400 Hz ±5 percent, phase C to neutral, shall be applied between pins n and A of J362.

Amplifier Characteristics

Input impedance	5K ohms min.
Load impedance	100 ohms ±3 percent +j100 ohms ±3 percent
Signal source impedance	1K ohm ±5 percent

Input Voltage	Output Voltage	Phase Relation
0.0	1.7V rms max	
0.060 ±0.001V rms	10.0 ±0.5V rms	±5°
0.350 ±0.004V rms	20.0 ±5V rms	±7.5°

4.3.1.1.8 A voltage and frequency of 26V rms ±3 percent 0°, 400 Hz ±5 percent, phase C to neutral, shall be applied across pins t and A (pin A neutral) of J362.

4.3.1.1.9 Roll servo amplifier and connections. The roll servo amplifier and connections shall be as follows:

Gyro Reference		Lear Amplifier No. 132056-01, or equivalent
J362 f	to	Pin 6
J361 h	to	Pin 5
J362 q	to	Pin 4
J362 A	to	Pins 7 and 3
J362 D	to	Pin 2
J362 g to J361 g		

A voltage and frequency of 26V rms ±3 percent +90°, 400 Hz ±5 percent, phase C to neutral, shall be connected between pins r and A of J362.

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Amplifier Characteristics

Input impedance	5K ohms min.
Load impedance	43 ohms ±1 percent +J63 ohms ±1 percent
Signal source impedance	1K ohm ±5 percent

Input Voltage	Output Voltage	Phase Relation
0	0.8 max	- - - - - - -
0.1 ±0.001V rms	21 ±1.0V rms	180° ±5°
0.35 ±0.002V rms	35 ±5V rms	- - - - - - -

4.3.1.1.10 A voltage and frequency of 18V rms ±3 percent 0°, 400 Hz ±5 percent, phase C to neutral, shall be applied across pins p and A (pin A neutral) of J362.

4.3.1.1.11 Leveling amplifier and connections. The leveling amplifier and connections shall be as follows:

Gyro Reference		Lear Amplifier No. 131884-01, or equivalent
J362 F	to	Pin 3
J362 i	to	Pin 6
J362 j	to	Pin 10
J362 A	to	Pins 4 and 8
J362 H	to	Pin 1
J362 G	to	Pin 5
J362 D	to	Pin 9

A voltage and frequency of 32V ±3 percent rms +90°, 400 Hz ±5 percent, phase C to neutral, shall be connected between pins J and A of J362.

Amplifier Characteristics

Input impedance	5K ohms min.
Load impedance	465 ohms ±3 percent +j697 ohms ±3 percent

Input Voltage dc	Output Polarity Pin i	Pin j	Output Volts rms	Phase	
0.050 ±0.001	+	-	10.5 ±3	180°	±5°
0.050 ±0.001	-	+	10.5 ±3	0°	±5°
0.25 ±0.001	+	-	24 ±2	180°	10.5°
0.25 ±0.001	-	+	24 ±2	0°	10.5°

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4.3.1.1.12 A voltage and frequency of 15V rms ±3 percent, 400 Hz ±5 percent, phase C to neutral, shall be applied across pins c and A of J362, with pins d and a of J362 connected and pins e and b of J362 connected.

4.3.1.1.13 A voltage and frequency of 15V rms ±3 percent, 400 Hz ±5 percent, phase C to neutral, shall be applied across pins R and A of J362, with pins M and K of J362 connected, pins N and L of J362 connected, and pins R and P of J362 connected.

4.3.1.1.14 These connections shall be the same as 4.3.1.1.1, except that pins C and B of J362 shall be disconnected and a 200-ohm ±5 percent 10w resistor inserted in series with pin C and a 680-ohm ±5 percent 10w resistor inserted in series with pin B.

4.3.1.1.15 These connections shall be the same as 4.3.1.1.7, except that 32V excitation to pins n and A of J362 shall not apply.

4.3.1.1.16 These connections shall be the same as 4.3.1.1.11, except that 32V excitation to pins J and A of J362 shall not apply.

4.3.1.1.17 A voltage and frequency of 50V rms ±3 percent, 400 Hz ±5 percent, phase C to neutral, shall be applied across pins R and A of J362, with pins M and K of J362 connected, pins N and L of J362 connected, and pins R and P of J362 connected.

4.3.1.1.18 A voltage and frequency of 50V rms ±3 percent, 400 Hz ±5 percent, phase C to neutral, shall be applied across pins c and A of J362 with pins d and a of J362 connected and pins e and b of J362 connected.

4.3.1.1.19 A voltage and frequency of 24.5 ±3 percent, 400 Hz ±5 percent, phase C, shall be applied between pins U and A of J362, and 19.3 ±1V rms, 400 Hz ±5 percent, phase C -90°, shall be connected between pins V and A of J362.

4.3.1.1.20 A voltage and frequency of 24.5 ±3 percent, 400 Hz ±5 percent, phase C ±180°, shall be applied between pins U and A of J362 and 19.3 ±1V rms, 400 Hz ±5 percent, phase C -90°, shall be connected between pins V and A of J362.

4.3.1.1.21 These connections shall be the same as 4.3.1.1.9, except for the following:

a. Pin X of J361 shall be connected to pin 5 of amplifier, Lear part No. 132056-01, or equivalent

b. Pin Y of J361 shall be connected to pin g of J362.

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4.3.1.1.22 Starting cycle. Unless otherwise specified, the displacement gyroscope assembly shall be started and the power specified under 4.3.1.1 shall be applied in accordance with the following procedure:

4.3.1.1.22.1 From 0 to 12 ±3 seconds time, power shall be applied as specified in 4.3.1.1.2 through 4.3.1.1.6, 4.3.1.1.8, 4.3.1.1.10, 4.3.1.1.14 through 4.3.1.1.17, and 4.3.1.1.21.

4.3.1.1.22.2 From 12 ±3 to 60 +20, -8 seconds time, power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.8, 4.3.1.1.10, 4.3.1.1.11, 4.3.1.1.17, 4.3.1.1.18, and 4.3.1.1.21.

4.3.1.1.22.3 After 60 +20, -8 seconds, power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13.

4.3.1.1.23 A voltage and frequency of 15V rms ±3 percent, 400 Hz ±5 percent, phase C +90° to neutral, shall be applied to pins E and F or B and C of J363 as specified under 4.6.

4.3.1.2 Amplifier-power supply

4.3.1.2.1 Phases A, B, C of the power defined under 3.13 shall be connected to pins B, C , and D, respectively, of J202 with pin A neutral. Line-to-neutral voltage shall be 115 ±2V rms. Line-to-line voltage shall be 200 ±4V rms.

4.3.1.2.2 A voltage of 26 ±3V dc shall be connected to pin K of J201.

4.3.1.2.3 A voltage of 14 ±0.5V rms, phase C, shall be connected (see 4.6.32.1).

4.3.1.2.4 A voltage of 45 ±1.0V rms, phase C, shall be connected (see 4.6.32.1).

4.3.1.2.5 Pitch signal. An isolated signal, variable from 0 to 0.35V rms and having a source impedance of 1,000 ohms ±5 percent, shall be connected between pins W and h of J201. Signal voltage shall be measured at the source side of the source impedance.

4.3.1.2.6 Leveling signal. An isolated, polarity reversible signal, variable from 0 to 0.25V dc, with a ripple voltage not exceeding 0.003V rms, shall be connected between pins j and i of J201.

4.3.1.2.7 Roll signal. An isolated signal, variable from 0 to 0.35V rms and with a source impedance of 1,000 ohms ±5 percent, shall be connected between pins f and e of J201. Signal voltage shall be measured at the source side of the source impedance.

4.3.1.2.8 Pitch. A load of 100 ohms ±3 percent +j100 ohms ±3 percent shall be connected between pins m and A of J201.

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4.3.1.2.8.1 A load of 100 ohms ±3 percent +3 percent +j100 ohms ±3 percent shall be connected between pins n and A of J201.

4.3.1.2.8.2 A load of 54 ohms ±3 percent +j67 ohms ±3 percent shall be connected between pins p and A of J201.

4.3.1.2.8.3 A load of 350 ohms ±3 percent +j30 ohms ±3 percent shall be connected between pins R and A of J201.

4.3.1.2.9 Leveling. A load of 465 ohms ±3 percent +j697 ohms ±3 percent shall be connected between pins H and A of J201.

4.3.1.2.9.1 A load of 700 ohms ±3 percent +j1150 ohms ±3 percent shall be connected between pins J and A of J201.

4.3.1.2.10 Leveling (10 ohms). A load of 10 ohms ±2 percent, 5w shall be connected between pins G and A ofJ201.

4.3.1.2.11 Roll. A load of 43 ohms ±3 percent +j63 ohms ±3 percent shall be connected between pins q and A of J201.

4.3.1.2.11.1 A load of 64 ohms ±3 percent +j90 ohms ±3 percent shall be connected between pins X and A of J201.

4.3.1.2.11.2 A load of 64 ohms ±3 percent +j90 ohms ±3 percent shall be connected between pins r and A of J201.

4.3.1.2.11.3 A load of 82 ohms ±3 percent +j154 ohms ±3 percent shall be connected between pins t and A of J201.

4.3.1.2.11.4 A load of 350 ohms ±3 percent +j30 ohms ±3 percent shall be connected between pins c and A of J201.

4.3.1.2.12 Starter unit. A load of 2,000 ohms ±5 percent, 10w resistors shall be connected between pins B and A and between pins C and A of J201.

4.3.1.2.13 Roll servo alarm. A signal of .58 ±.02V ac shall be connected from terminal e of J201 to terminal f of J201. Terminal E shall be connected to terminal F of J202.

4.3.1.2.14 Pitch servo alarm. A signal of 1.70 ±.05V ac shall be connected from terminal U to terminal h of J201.

4.3.1.3 Compass adapter. Connections shown on figure 8 for the specified loads, or equivalent connections shall be made.

4.3.1.3.1 Flux valve. A flux valve in accordance with MIL-D-25521, or equivalent, shall be provided and connected to the adapter as shown on figure 8.

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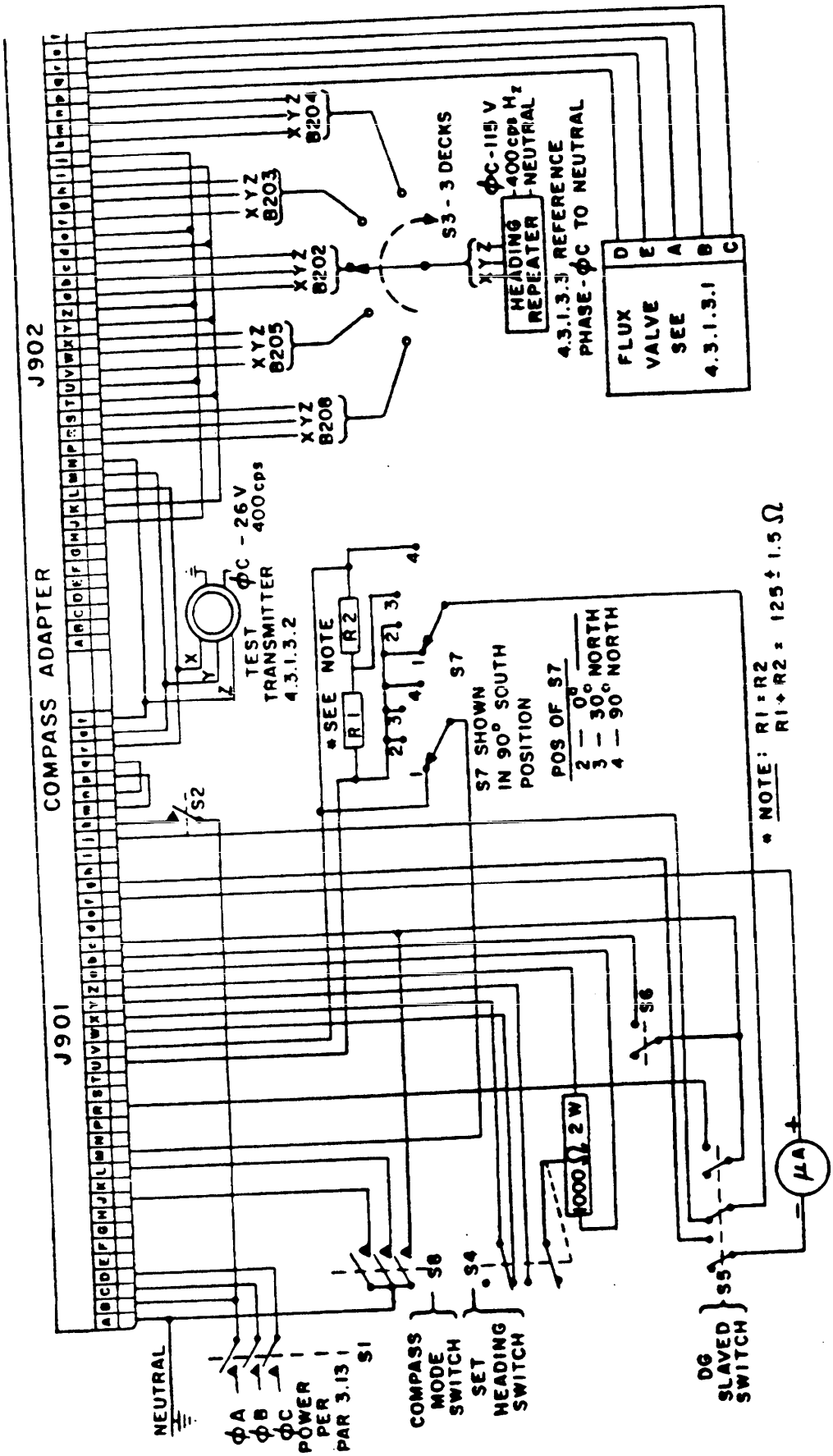


FIGURE 8. Switch Arrangements

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4.3.1.3.2 Test transmitter. The test transmitter shall be a Kearfott type R510-1A, or equivalent, properly calibrated. It shall be possible to read the shaft position within 0° 6'. The transmitter shall be nulled as specified in MIL-S-16892. The three stator leads shall be designated X, Y, and Z, and the rotor leads C and H. At 0° position with Z and C to neutral and H to 26V ±2 percent 400 ±4 Hz, phase C, the voltage across X and Y shall be a minimum and when the shaft is turned in an increasing heading direction, the voltage across X and Z shall increase before decreasing and the voltage across Y and Z shall decrease before increasing. The voltage across H and X or Y shall be greater than the rotor excitation voltage. The test transmitter shall be connected to the compass adapter as shown on figure 8.

4.3.1.3.3 Heading repeater. The heading repeater shall be an Eclipse-Pioneer type 1-3647-1-A, or equivalent. Power applied to the heading repeater shall be 115 ±4V 400 ±10 Hz, and connections shall be made as indicated on figure 8.

4.3.1.3.4 Miscellaneous. Switch arrangements shall be made in accordance with figure 8.

4.3.1.4 Compass controller

4.3.1.4.1 Indicator minus. A current of 50 ±2.5 microamperes (ua) dc shall be connected to pin J801-a with respect to pin J801-b.

4.3.1.4.2 Indicator plus. This condition shall be the same as in 4.3.1.4.1 except polarity shall be reversed.

4.3.1.4.3 Lighting power. A voltage of 28 ±0.5V dc shall be connected to pin J801-j with respect to J801-A.

4.3.1.5 Control assembly

4.3.1.5.1 Power conditions. Power as defined under 3.13 shall be applied as shown on figure 9.

4.3.1.5.2 Synchro repeater. The synchro outputs shall be read with an Eclipse-Pioneer type 1-3647-1-A, or equivalent.

4.3.1.5.3 Flux valve. The DT-173/AJN detector conforming to MIL-D-25521, or equivalent, shall be provided and connected as shown on figure 9.

4.3.1.5.4 Interconnections. The control assembly components shall be connected together as shown on figure 9.

4.3.2 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C).

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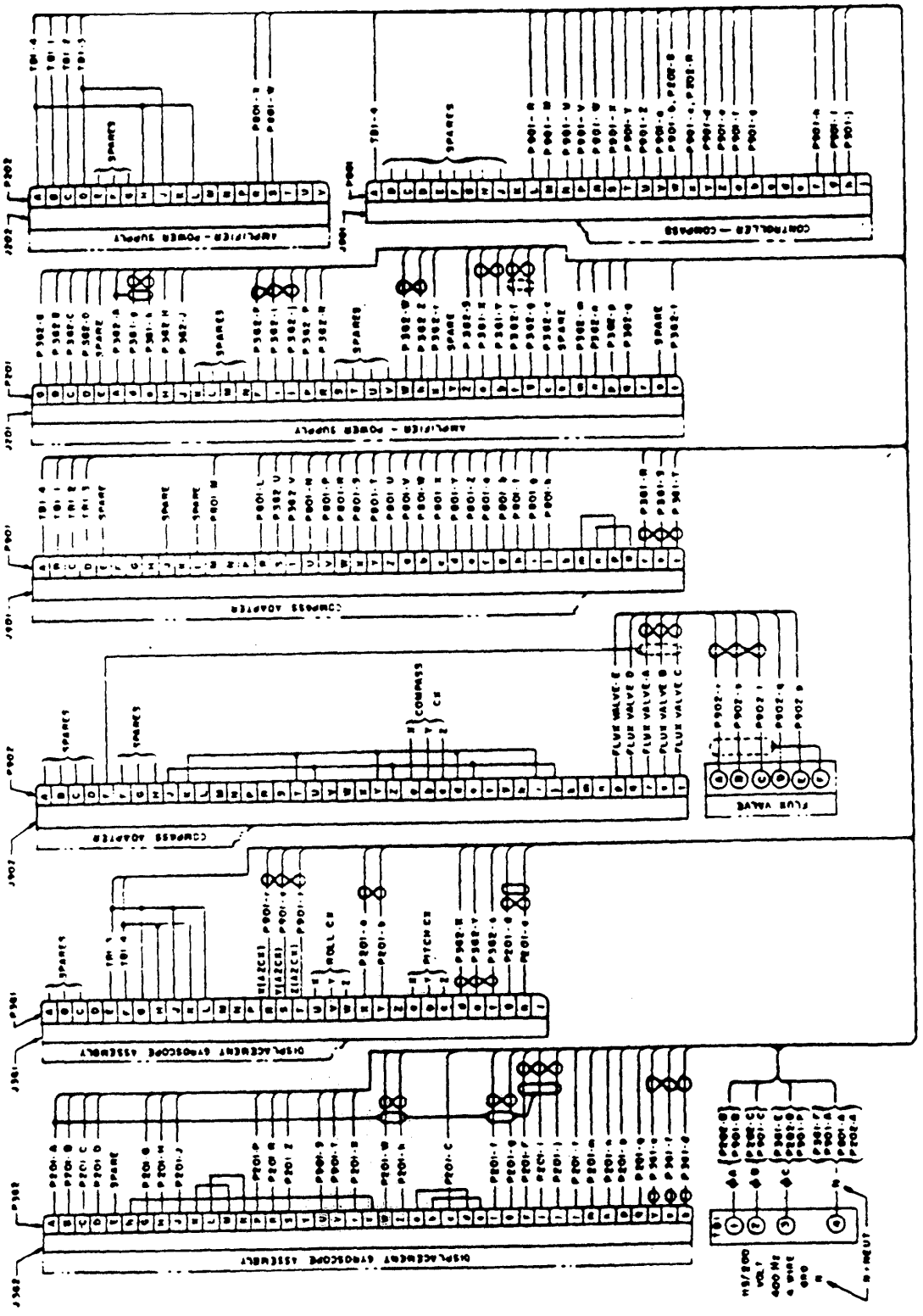


FIGURE 9. Interconnections

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4.3.3 Attitude. Unless otherwise specified, the control assembly components shall be tested in the normal operating position. The displacement gyroscope assembly shall be positioned in one of the following attitudes, as specified for the applicable test:

a. The gyroscope assembly standing on a surface which is level within 0° 6 .

b. The gyroscope assembly mechanically secured to a surface which is level within 0° 6′.

c. The gyroscope assembly mounted on and mechanically secured to a leveled (within 1/4°) scorsby table and the scorsby oscillating simultaneously in three mutually perpendicular planes (roll, pitch, and yaw) with the total movement in each plane of 2° at a rate of 5 to 7 oscillations per minute. The accelerating type scorsby table shall be reversed once per minute.

d. The gyroscope assembly mounted on a fixture permitting 360° rotation about a horizontal axis so that the pitch axis of the vertical gyro is parallel to the fixture axis.

e. Same as attitude e, except the roll axis of the vertical gyro shall be parallel to the fixture axis.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of seven control assemblies representative of the production equipment. Three of the control assemblies shall be subjected to all the tests except reliability and the other four shall be subjected to the reliability test only. The samples, identified with the manufacturer's part number and such other information as required, shall be submitted to the procuring activity.

4.4.1.1 Reliability demonstration. The four control assemblies specified in 4.4.1 shall be subjected to the test specified in 4.6.86.1 for a total of 6,240 hours unless rejection is determined in accordance with figure 10 prior to that time.

4.4.2 Preproduction test report. After the contractor completes the preproduction tests, he shall prepare a preproduction test report in accordance with MIL-STD-831 and furnish two complete copies of the report to the procuring activity.

4.4.3 Preproduction tests. Preproduction tests shall consist of all the tests specified under 4.6 except 4.6.86.2.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests.

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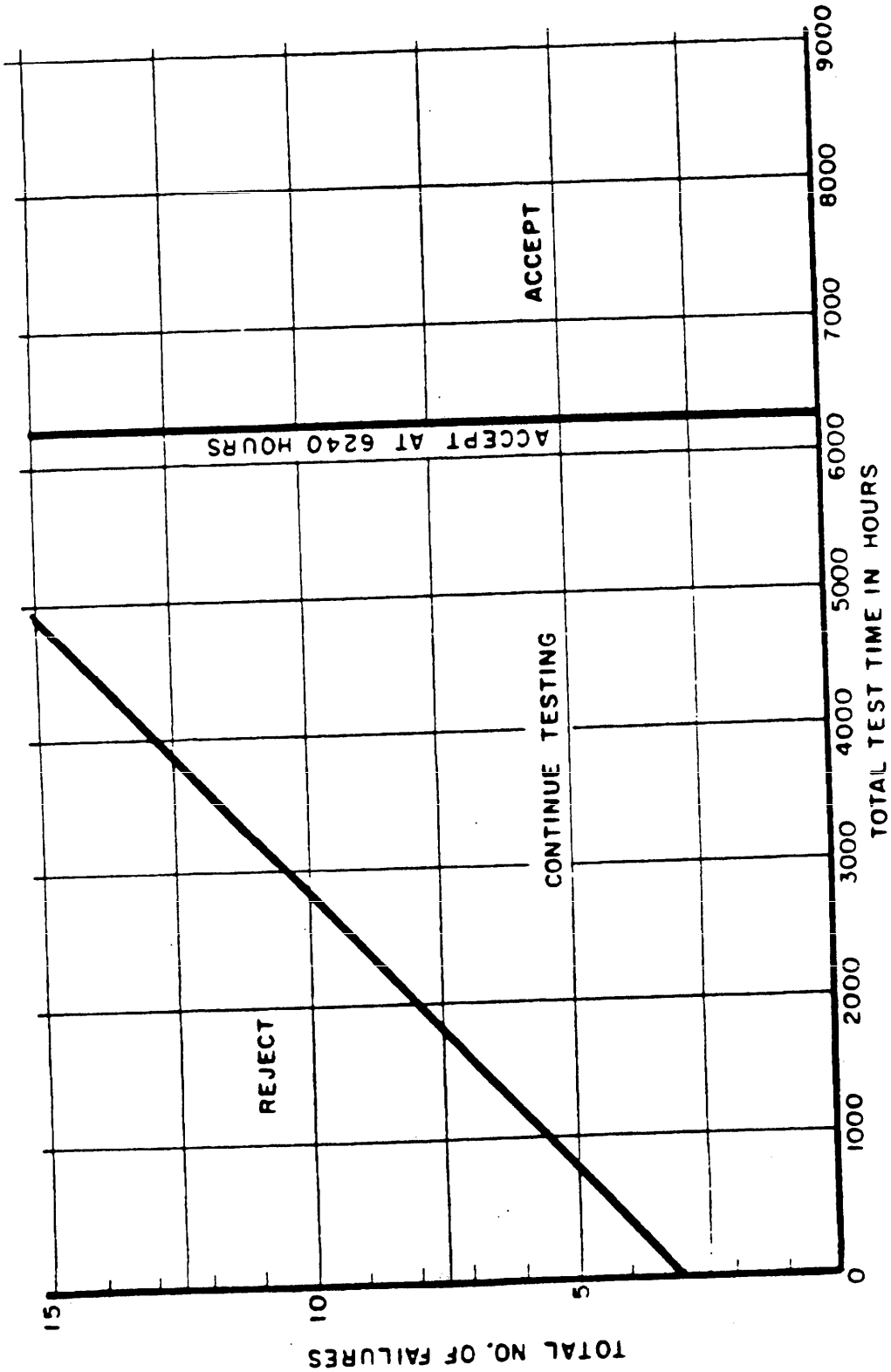


FIGURE 10. Reject - Continue Test Criteria

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4.5.1 Individual tests. Each component of the control assembly shall be subjected to the following applicable tests as described under 4.6:

a. DISPLACEMENT GYROSCOPE ASSEMBLY

(1) Examination of product

(2) Initial erection

(3) Final erection

(4) Verticality

(5) Starting time

(6) Slip ring continuity

(7) Synchro phasing and zeroing

(8) Synchro null coincidence

(9) Synchro null voltage

(10) Slaving

(11) Leveling

(12) Fast erection

(13) Slow erection

(14) Erection cutout

(15) V. G. correction, roll

(16) V. G. correction, pitch

(17) Roll, pitch, and yaw

(18) Loop test

(19) Gimbal follow-up

(20) Power interruption - gyroscope

(21) Latitude correction - gyroscope

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(22) Vertical gyro drift

(23) Directional gyro drift

(24) Maximum drift

(25) Random drift

(26) Mean drift

(27) Roll switching

(28) Pitch switching

(29) Power failure

(30) System compatibility

(31) Case leakage.

b. AMPLIFIER-POWER SUPPLY

(1) Examination of product

(2) Relay operation

(3) Roll erection

(4) Pitch erection cutout

(5) Fast erection power supply

(6) Power interruption - amplifier - power supply

(7) Failure warning

(8) Pitch amplifier

(9) Leveling amplifier

(10) Roll amplifier

(11) System compatibility.

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c. COMPASS ADAPTER

(1) Examination of product

(2) Azimuth servo operation

(3) Azimuth followup rate

(4) Azimuth slaving rate and cutout

(5) Latitude correction

(6) Polarity

(7) Annunciator

(8) Deviation compensation

(9) Static settling

(10) Slaving accuracy

(11) Compass mode accuracy

(12) Azimuth servo accuracy

(13) Bias correction

(14) Automatic gain control

(15) Dynamic accuracy

(16) Manual synchronization

(17) Switchover synchronization

(18) Push to sync operation

(19) Heading set slew

(20) Autopilot slewing decoupling

(21) Synchro zeroing and sensing

(22) System compatibility.

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d. COMPASS CONTROLLER

(1) Examination of product

(2) Synchronizing indicator

(3) Mode selector switch

(4) Set heading control

(5) Latitude control

(6) Push to synchronization

(7) Lighting

(8) System compatibility.

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. One component shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as applicable and as described under 4.6:

a. Individual tests

b. Voltage and frequency variation

c. Low temperature operation

d. High temperature operation

e. Dielectric strength

f. Vibration error

g. Low field strength

h. Balance

4.5.2.2 Sampling plan B. Unless otherwise specified, 3 samples of each component shall be selected at random from the first 15 items on the contract or order and subjected to the following tests, as applicable, in the sequence listed and as described under 4.6:

a. Sampling plan A tests

b. Radio noise interference and susceptibility

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c. Acceleration

d. Temperature-altitude

e. Temperature shock

f. Shock

g. Humidity

h. Fungus

i. Rain

j. Salt fog

k. Dust

l. Explosive atmosphere

m. Acoustical noise

n. Sealing thermal shock

o. Vibration failure

p. Vibration endurance.

4.5.2.2.1 At the option of the procuring activity, the tests may be divided into two groups as follows, to be conducted on two sets of three instruments each, if a need exists to expedite testing. The components shall meet the requirements of the sampling plan A tests before beginning the group of tests to which submitted. Group II tests, with the exception of the acoustical noise test, may be performed on hermetically sealed empty cases, with prior approval from the procuring activity. All external parts shall be installed or attached.

Group I	Group II
Radio noise interference and susceptibility	Humidity
Acceleration	Fungus
Temperature-altitude	Rain
Temperature shock	Salt fog
Shock	Dust
Vibration failure	Explosive atmosphere
Vibration endurance	Acoustical noise
	Sealing thermal shock

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4.5.2.2.2 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.5.2.2.3 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.2.2.4 Corrective action and retest. Corrective action and retest shall be required for each failure that can be reasonably attributed to poor manufacture or design practices or that adversely affects mission completion.

4.5.2.3 Sampling plan C. Unless otherwise specified, 3 control assemblies shall be selected at random from each 50 produced on the contract or order and subjected to the test specified in 4.6.86.2.

4.5.2.3.1 Acceptance or rejection. Figure 11 shall be the criteria used to determine whether the equipment will be accepted or rejected. Certified time records shall be maintained of the test time accumulated on the equipment after starting the environmental cycling specified in 4.6.86. In the event a continue-test decision applies after 1,800 hours of testing, the production quantity shall be considered as conditionally acceptable; however, the accumulated time and failures shall be added to the data of the next production quantity for the composite accept-reject decision. If the continue-test condition persists, the data from successive production quantities shall be accumulated until 15 failures have occurred or 6,240 hours of testing have been completed at which time an accept-reject decision shall be made.

4.5.2.4 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The control assembly components shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Initial erection. After completion of 4.3.1.1.22.2, the pitch synchro output signals at pins a, b, and c of J361 and the roll synchro output signals at pins X, Y, and Z of J361 shall not exceed 1° with the gyroscope positioned in accordance with 4.3.3b.

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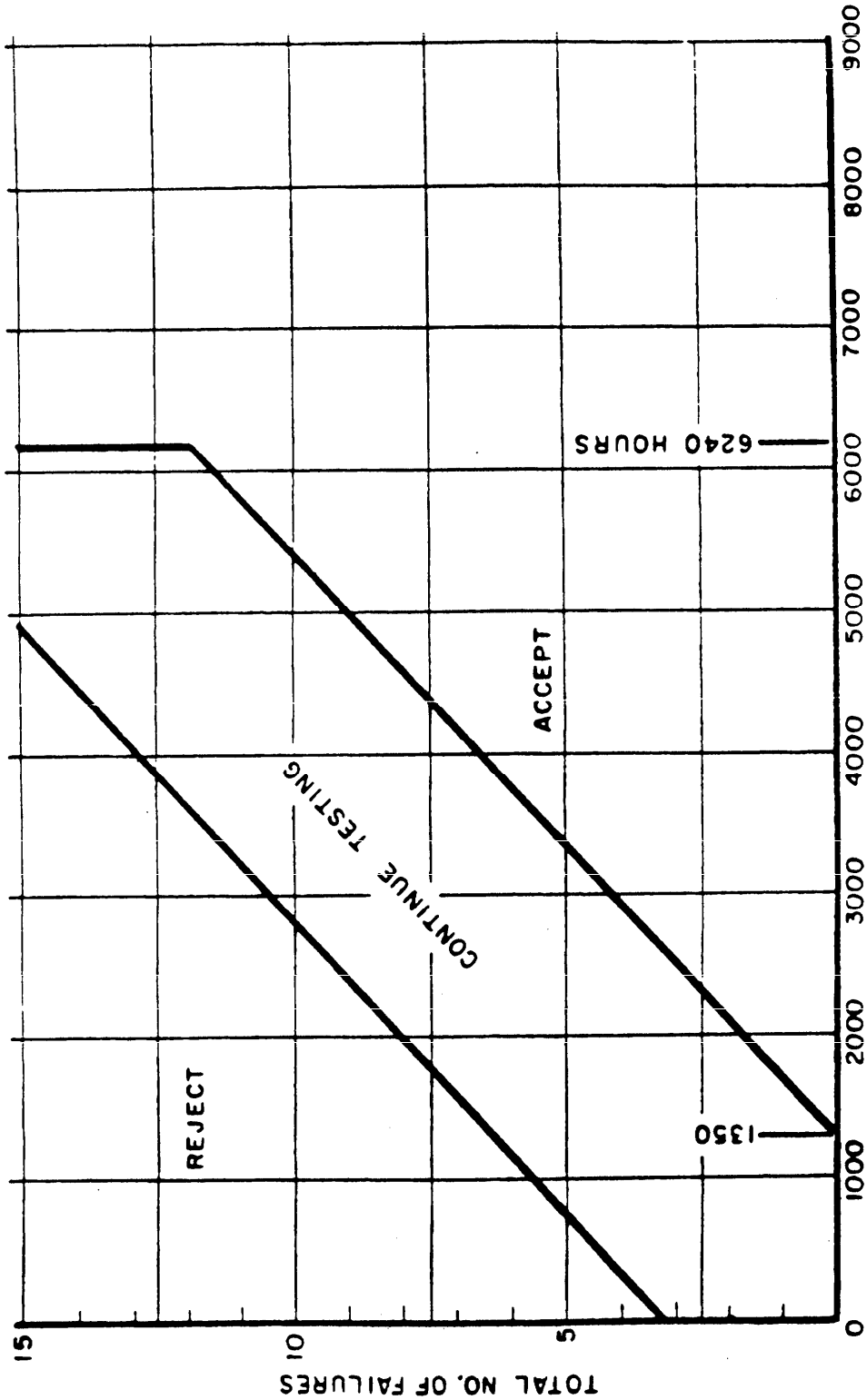


FIGURE 11. Accept - Continue Test - Reject Criteria

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4.6.3 Final erection. Within 2 minutes after completion of 4.3.1.1.22.2, the pitch synchro output at pins a, b, and c, and at pins d, e, and f of J361 shall not exceed 0.5°. The roll synchro output signal at pins X, Y, and Z, and at pins U, V, and W of J361 shall not exceed 0.75°. A low-angle scorsby may be used during the final erection period while positioned in accordance with 4.3.3b.

4.6.4 Verticality. After completion of 4.3.1.1.22.2, power shall be disconnected as specified in 4.3.1.1.12 and 4.3.1.1.13. Rotation of the reference 180° in azimuth shall not cause the vertical gyro to change more than 0.5° in either roll or pitch. The vertical gyro shall meet this requirement from any of the four cardinal headings after allowing a minimum of 2 minutes on low erection. A low-angle scorsby shall be used during the 2-minute low-erection period.

4.6.5 Starting time. After application of power as specified in 4.3.1.1.1, the gyro rotors shall attain full speed from rest in 5 minutes or less.

4.6.6 Slipring continuity. Power shall be applied in accordance with 4.3.1.1.1 through 4.3.1.1.13 and the current in each gyroscope motor phase shall have reached a stable minimum.

4.6.6.1 Under the conditions listed below, no sharp peaks or discontinuities shall occur in the azimuth synchro output signals at pins R, S, and T of J361, and pins M, N, and P of J361 when loaded with an Eclipse-Pioneer autosyn servo test indicator type 1-3647-1-A, or equal.

a. Rotating the reference case approximately 2 rpm in azimuth

b. Tilting the reference case approximately ±80° about a horizontal axis perpendicular to its fore-and-aft axis

c. Tilting the reference case approximately 2 rpm about its fore-and-aft axis.

4.6.6.2 Under the following conditions, no sharp peaks or discontinuities shall occur in the pitch synchro output singals at pins a, b, and c of J361, pins W and h of J362, and pins d, e, and f of J361 when loaded with an Eclipse-Pioneer autosyn servo test indicator type 1-3647-1-A, or equal.

a. Tilting the reference case approximately 2 rpm about its fore-and-aft axis

b. Tilting the reference case approximately ±80° about a horizontal axis perpendicular to its fore-and-aft axis.

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4.6.6.3 Under the following conditions, no sharp peaks or discontinuities shall occur in the roll signals at pins X, Y, and Z, pins U, V, and W, and pins g, h, and J of J361, when loaded with an Eclipse-Pioneer autosyn servo test indicator type 1-3647-1-A, or equal:

a. Rotating the gyro case approximately 1/2 rpm about its fore-and-aft axis

b. Power connected as in 4.6.6, except 4.3.1.1.21 shall apply and the gyro case rotated approximately 2 rpm about a horizontal axis perpendicular to its fore-and-aft axis

c. Tilting the reference case approximately ±80° about its fore-and-aft axis.

4.6.7 Synchro phasing and zeroing. Power shall be supplied as specified in 4.3.1.1.1 through 4.3.1.1.13. Attitude shall be in accordance with 4.3.3b.

4.6.7.1 Directional gyro, azimuth synchros. The case shall be oriented in azimuth so that turning the reference case in a cw direction in a horizontal plane will produce the following:

a. Bottom synchro - An output voltage change between pins M to N of J361 which is out of phase with the rotor excitation voltage pins G to F of J361. (Stator voltage between pins M and P of J361 shall be out of phase with pins G to F of J361.)

b. Top synchro - An output voltage change between pins R and S of J361 which is out of phase with the rotor excitation voltage pins E of J361 to A of J362. (Stator voltage between pins R and T of J361 shall be out of phase with pin E of J361 to pin A of J362.)

4.6.7.1.1 Leveling pick-off. With the power specified in 4.3.1.1.21 replacing the power specified in 4.3.1.1.9 and the case rotated 90° cw in azimuth, from the position established in 4.6.7.1b, tilting the reference case about its fore-and-aft axis in a direction to simulate a left bank shall produce d-c output voltage between pins i to A of J362 with pin i positive.

4.6.7.2 Synchro phasing, vertical gyro. Tilting the reference case to simulate a dive shall produce the following:

a. Pitch output synchro - An output voltage change between pins a to b of J361 which is out of phase with the synchro excitation voltage between pins L and K of J361. (Stator voltage between pins a and c of J361 shall be in phase with the rotor excitation voltage between pins L and K of J361.)

b. Pitch servo synchro - An output voltage change between pins d and e of J361 which is out of phase with the synchro excitation voltage between pins D and A of J362. (Stator voltage between pins d and f of J361 shall be in phase with rotor excitation voltage between pins D and A of J362.)

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4.6.7.2.1 Inner roll synchro. With the power specified in 4.3.1.1.21 replacing the power in 4.3.1.1.9, tilting the reference case about its fore-and-aft axis in a direction to simulate a left bank shall produce an inner roll synchro output voltage change between pins g and h of J361 which is out of phase with the inner roll synchro excitation voltage between pins D and A of J362. (Stator voltage between pins g and j of J361 shall be in phase with the rotor excitation voltage between pins D and A of J362.)

4.6.7.3 Outer roll synchros. Tilting the reference case about its fore-and-aft axis in a direction to simulate a left bank shall produce the following:

a. Outer roll synchro - VG end - An output voltage change between pins X and Y of J361 which is out of phase with the outer roll synchro excitation voltage between pins D and A of J362. (Stator voltage between pins X and Z of J361 shall be in phase with the rotor excitation voltage between pins D and A of J362.)

b. Outer roll synchro - DG end - An output voltage change between pins U and V of J361 which is out of phase with the outer roll synchro excitation voltage between pins J and H of J361. (Stator voltage between pins U and W of J361 shall be in phase with the rotor excitation voltage between pins J and H of J361.)

4.6.8 Synchro null coincidence. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13 and attitude shall conform to 4.3.3b.

4.6.8.1 Azimuth. When the synchro output voltages at pins M, N, and P of J361 indicate null, the synchro output voltages at pins R, S, and T of J361 shall indicate within the equivalent of 0.4° from null.

4.6.8.2 Pitch. When the synchro output voltages at pins a, b, and c of J361 indicate null, the synchro output voltages at pins d, e, and f of J361 shall indicate within the equivalent of 0.4° from null.

4.6.8.3 Roll. When the synchro output voltages at pins U, V, and W of J361 indicate null, the synchro output voltages at pins X, Y, and Z of J361 shall indicate within the equivalent of 0.4° from null.

4.6.8.4 Inner outer roll. When the inner roll synchro output voltages at pins g, h, and j of J361 indicate null, the outer roll synchro output voltages at pins X, Y, and Z of J361 shall indicate within the equivalent of 0.7° from null when the outer roll gimbal is set at the verticality point.

4.6.9 Synchro null voltage. The null voltage of the following synchros shall be measured with an impedance of 100 +j570 ohms and shall not exceed 30 mv rms:

a. The azimuth synchro between pins R and S of J361

b. The azimuth synchro between pins M and N of J361

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c. The inner roll synchro between pins g and h of J361

d. The pitch synchro between pins a and b of J361

e. The pitch servo synchro between pins d and e of J361

f. The outer roll synchro between pins X and Y of J361

g. The outer roll synchro between pins U and V of J361.

4.6.10 Slaving. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13 and attitude shall conform to 4.3.3d. When the current in each gyroscope rotor phase has attained a stable minimum, application of power specified in 4.3.1.1.19 and 4.3.1.1.20 shall produce a slaving rate in ccw and cw directions between 1° and 2° per minute after correction for the earth's rate.

4.6.11 Leveling. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13 and attitude shall conform to 4.3.3d. When the current in each gyroscope rotor phase has attained a stable minimum, the leveling axis shall be displaced from null, the computed rate of the leveling axis movement for a minimum of 5° shall be between 2° and 6° per minute. The rate shall be taken beyond the saturation point of the amplifier.

4.6.12 Fast erection. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.11 and the initial attitude shall conform to 4.3.3.c. With the current in each gyroscope rotor phase at a stable minimum, application of power specified in 4.3.1.1.18 and 4.3.1.1.17 shall erect the roll and pitch axes, respectively, from an approximate 15° position either side of vertical at a minimum rate of 15° per minute, except at extreme temperature where the minimum rate shall be 13° per minute. The pitch and roll axis fast erection rates shall be determined separately.

4.6.13 Slow erection. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13, attitude shall conform to 4.3.3d, and the current in each gyroscope rotor phase shall have attained a stable minimum.

4.6.13.1 Pitch. With the terminal end of the gyroscope assembly oriented to either compass north or south, the vertical gyro shall erect 5° from an approximate 10°-dive position at a rate of 0.8° to 1.8° per minute. The vertical gyro shall also erect 5° from an approximate 10°-climb position at a rate within the above limits. During these tests, the undeflected (roll) axis shall not deviate from vertical more than 1°. With the power specified in 4.3.1.1.13 disconnected, there shall be no erection in a simulated dive or climb position.

4.6.13.2 Roll. With the terminal end of the reference oriented to either compass east or west, the vertical gyro shall erect 5° from an approximate 10° right bank position at a rate of 0.8° to 1.8° per minute. The vertical gyro shall also erect 5° from an approximate 10° left bank position at a rate within the above limits. During

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the erection tests, the undeflected (pitch) axis shall not deviate from vertical more than 1°. With the power specified in 4.3.1.1.12 disconnected, there shall be no erection in a simulated right or left bank position.

4.6.14 Erection cutout. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13. The vertical gyro shall then be displaced approximately 15° in right bank and either direction in pitch. Power shall be removed from pin c of J362 and the gyro shall continue erecting in pitch but shall cease to erect in bank. Power shall be reconnected to pin c of J362 and power removed from pin P of J362. The gyro shall begin erecting in bank but cease to erect in pitch. This test shall be repeated with the deflections in the opposite directions.

4.6.15 VG correction, roll. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13 and attitude shall conform to 4.3.3d. With the power specified in 4.3.1.1.12 disconnected, the signal specified in 4.3.1.1.23 shall be applied to pins E to F of J363 (in phase with phase C +90° to neutral). The resulting roll precession shall be in left bank at a rate of 1 ±0.15° per minute after proper correction for the earth's rotation. The procedure shall be repeated, except the signal specified in 4.3.1.1.23 shall be applied to pins F to E (in phase with phase C +90° to neutral). The resulting roll precession shall be in right bank at a rate of 1 ±0.15° per minute after proper correction for the earth's rotation.

4.6.16 VG correction, pitch. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13 and attitude shall conform to 4.3.3d. With the power specified in 4.3.1.1.13 disconnected, the signal specified in 4.3.1.1.23 shall be applied to pins B to C of J363 (in phase with phase C +90° to neutral). The resulting pitch precession shall be in dive at a rate of 1 ±0.15° per minute after proper correction for the earth's rotation. The procedure shall be repeated, except the signal specified in 4.3.1.1.23 shall be applied to pins C to B of J363 (in phase with phase C +90° to neutral). The resulting precession shall be in climb at a rate of 1 ±0.15° per minute after proper correction for the earth's rotation.

4.6.17 Roll, pitch, and yaw. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13 and attitude shall conform to 4.3.3d, except total movement shall be 15° in each plane. After 15 minutes of operation, the scorsby shall be releveled and the indication immediately read. The vertical gyro shall be within 1/2° of the local gravity vertical.

4.6.18 Loop test. With the gyroscope assembly mounted as specified in 4.3.3e, the following tests shall be conducted:

4.6.18.1 Normal operation. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13. The gyroscope assembly shall be rotated through one complete revolution (loop) at a rate of 30° per second. The gyroscope assembly shall be leveled immediately and displacement of the gyros noted. The vertical gyro shall be erected to within 1/2° of the local gravity vertical. The directional gyro shall not have deviated by more than 1/2°.

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4.6.18.2 Special operation. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.8, 4.3.1.1.10 through 4.3.1.1.13, and in 4.3.1.1.21. The gyroscope assembly shall be rotated through one complete revolution (loop) at a rate of 200° per second. The gyroscope assembly shall be leveled immediately and displacement of the gyro noted. The vertical gyro shall be erect to within 1/2° of the local gravity vertical. The directional gyro shall not have deviated more than ±1/4°.

4.6.18.3 Third gimbal operation. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.8, 4.3.1.1.10 through 4.3.1.1.13, and 4.3.1.1.21. The gyroscope assembly shall be displaced 90° ±2° in pitch and then displaced 15° to 22° in yaw. Application of power as specified in 4.3.1.1.1 through 4.3.1.1.13 shall cause the roll signal to change 90° ±5° and the pitch signal to change approximately 20°.

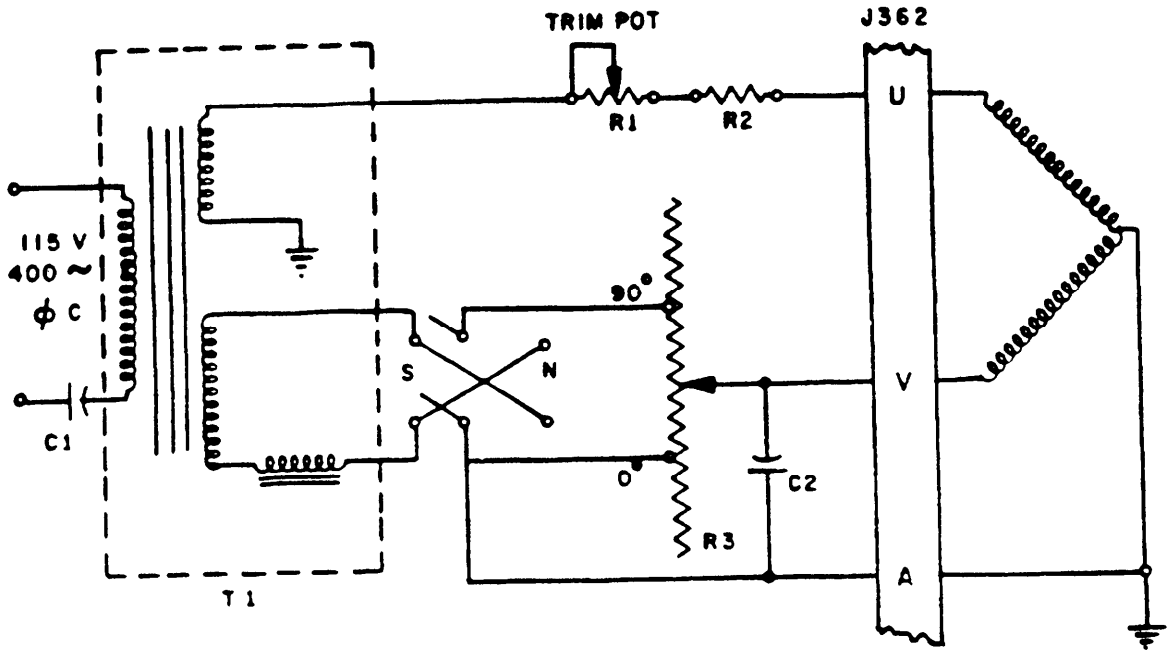
4.6.19 Gimbal followup. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13 and attitude shall conform to 4.3.3e. The gyroscope assembly shall be rotated in roll 360° per second. The servoed gimbal shall remain level within 20° and shall move smoothly with no hunting or jumping.

4.6.20 Power interruption-gyroscope. The gyroscope assembly shall be mounted as specified in 4.3.3e and 4.3.3f and power applied as specified in 4.3.1.1.1 through 4.3.1.1.13. After 5 minutes of operation, the power shall be removed and the gyroscope assembly moved through angles of ±60° in pitch and ±75° in roll and returned to level. Power shall be reapplied 40 seconds after removal. The pitch and roll axes shall indicate level within ±1°.

4.6.21 Latitude correction gyro. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13 and figure 12 or equivalent. Attitude shall conform to 4.3.3d. The trim pot shall be so adjusted that the voltage between pins U and A of J362 coincides with the recommended trim voltage labeled on the connector end of the gyro reference. The latitude correction device shall be set to the electrical equivalent of 0°, 60° north and 90° south, for a period of not less than 1/2 hour at each setting. The extremes of the output at pins R, S, and T shall be read and recorded for each period. The drift rate of the gyro shall not differ by more than ±2° per hour from the difference between the drift computed for the local latitude and the drift correction computed for the latitude set on the latitude correction control.

4.6.22 Vertical gyro drift. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.11, attitude shall conform to 4.3.3d, and the current in each phase of the gyroscope rotors shall have attained a stable minimum. With the spin axis deflected 20° ±0.5° in the easterly direction and the undeflected axis within ±0.5° of local gravity vertical, free drift measurements shall be taken for a 10-minute period with the gyro oriented approximately to each of the four cardinal headings. Drift of the deflected and undeflected axis shall be measured and the drift shall not exceed 15° per hour with proper correction for the earth's rotation.

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R1 = 1K W.W. 10W.
R2 = 500 ± 5% Ω WW3W
R3 = LEAR P/N 129677-01 OR EQUAL

C1 = 0.22 ± 5% UF 1000 WV
C2 = 0.068 ± 5% UF 200 WV
T1 = LEAR P/N 133325-01 OR EQUAL

A DIAL SHALL BE ATTACHED TO THE SHAFT OF R3 WITH A SCALE LABELED AS SHOWN BELOW

THE ELECTRICAL ZERO OF THE POTENTIOMETER SHALL BE ALIGNED TO THE MECHANICAL ZERO OF THE DIAL

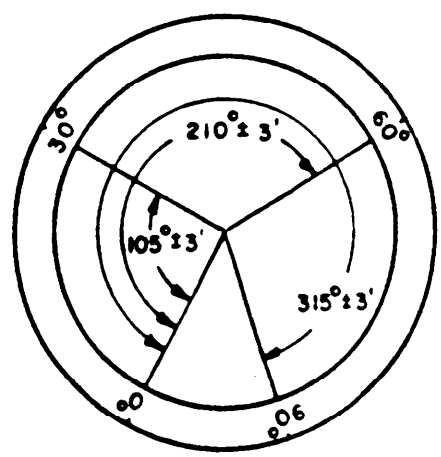


FIGURE 12. Schematic

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4.6.23 Directional gyro drift. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13 and the gyro shall be mounted in accordance with 4.3.3d. The output of pins R, S, and T of J361 shall be recorded within 15 minutes after application of power and each half hour thereafter for a total of 4 hours. Eight hourly drift rates, properly corrected for the earth's rotation, shall be computed from consecutive half-hour readings.

4.6.23.1 Maximum drift. The maximum hourly drift rate determined in 4.6.23 shall not exceed ±1.0° per hour.

4.6.23.2 Random drift. The normally distributed random drift of the directional gyro shall not exceed ±0.25° per hour - one sigma, as shown by compliance of the eight hourly drift rates of 4.2.23 with the formula:

$$0.25 \geq \sqrt{\frac{\Sigma (x_i - \bar{x})^2}{8}}$$

Where $\bar{X}$ is the mean drift rate and x_i represents each of the 8 hourly drift rates.

4.6.23.3 Mean drift during environmental extremes. The average of the eight hourly drift rates of 4.6.23 shall not exceed ±1.5° per hour.

4.6.24 Roll switching. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13 and attitude shall conform to 4.3.3b. When the reference is displaced in right and left bank 90° ±2° from its initial position, there shall be continuity between pin D of J361 and pin A of J362.

4.6.25 Pitch switching. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.8, 4.3.1.1.10 through 4.3.1.1.13, and in 4.3.1.1.21, and attitude shall conform to 4.3.3b. When the reference is displaced in dive and climb 90° ±2° from its initial position, there shall be continuity between pins T and A of J362.

4.6.26 Power failure. Power shall be applied as specified in 4.3.1.1.1 through 4.3.1.1.13 and the current in each phase of the gyroscope rotors shall have attained a stable minimum. Power shall be disconnected from pin B of J362. The voltage measured between pins S and A of J362 shall be a minimum of 15V ac. The test shall be repeated, disconnecting pins C and D of J362. The same tolerance shall apply.

4.6.27 Relay operation. Voltages, continuity, and time sequence shall be checked in accordance with the following after application of power as specified in 4.3.1.2.1. Starting time shall begin simultaneously with application of power. Phase reference voltage shall be phase C to neutral (pin D to pin A of J201). The voltages shall be

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measured with the unit loaded per 4.3.1.2.8 through 4.3.1.2.12. The time periods at the voltages and at the pins specified shall be as follows:

a. Time - 0 to 12 seconds

Pins of J201	Voltage Output RMS
B to A	86 ±7V
C to A	103 ±7V
H to A	2.0V max
M to A	2.5V max
R to A	50 ±5.5V
c to A	Approximately 0V (this voltage shall increase to 50 ±5.5V at 12 ±3 sec)
L(J202) to K(J202)	Open

b. Time - 12 to 60 seconds

Pins of J201	Voltage Output RMS or Continuity
B to A	115V ±3 percent
C to A	115V ±3 percent
c to A	50 ±5.5V
g to b	Continuity
R(J202) to S(J202)	Continuity (circuit shall open at 60 +20, -8 seconds)
L(J202) to K(J202)	Open

c. Time - After 60 seconds

Pins of J201	RMS Voltage or Continuity
B to A	115V ϕA ±3 percent
C to A	115V ϕB ±3 percent
D to A	115V ϕC ±3 percent
J to A	33 ±5V 90° ±5° leading
R to A	15 ±2.0V 0° ±5°
X to A	26.5 ±3.5V 90° ±5° leading
c to A	15 ±2.0V 0° ±5°
p to A	18.3 ±2.0V 0° ±5°
t to A	26.5 ±3.5V 0° ±5°
J to n	Continuity
R to P	Continuity
g to d	Continuity
t to r	Continuity

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A to A (J202)	Continuity
D to D (J202)	Continuity
C(J202) to P(J202)	Continuity
R(J202) to S(J202)	Open
L(J202) to K(J202)	Continuity

4.6.28 Roll erection cutout. With power applied as specified in 4.3.1.2.1 and pins N and P of J202 jumpered, there shall be no voltage between pins c and A of J201.

4.6.29 Pitch erection cutout. With power applied as specified in 4.3.1.2.1 and pins M and P of J202 jumpered, there shall be no voltage between pins R and A of J201.

4.6.30 Fast erection power supply. With power applied as specified in 4.3.1.2.1 and pins V and U of J202 jumpered, the voltage between pins c and A of J201 shall change from approximately 15 to approximately 50V.

4.6.31 Power interruption - amplifier-power supply. Following at least 5 minutes of continuous operation with power applied as specified in 4.3.1.2.1, all phases of the power shall be interrupted simultaneously for a period of 40 seconds, then reapplied. There shall be no continuity between pins K and L of J202 during the 40-second period. When power is reapplied, recycling shall not occur and continuity shall return between pins K and L.

4.6.32 Failure warning

4.6.32.1 Power. Power as specified in 4.3.1.2.1 shall be applied and the signal as specified in 4.3.1.2.4 shall be applied between pins Z of J201 and A of J202. When the unit is excited as above, continuity between pins L and K of J202 shall be broken. The signal of 4.3.1.2.4 shall be removed and the signal specified in 4.3.1.2.3 applied. Continuity between pins L and K of J202 shall be broken.

4.6.32.2 Roll servo alarm. Power as specified in 4.3.1.2.1 and 4.3.1.2.2 and signal specified in 4.3.1.2.13 shall be applied. Pins E and F of J202 shall be connected. There shall be continuity between pins L and K of J202. The power of 4.3.1.2.2 shall be removed and the continuity from pins L to K shall disappear. The signal of 4.3.1.2.13 shall be removed and the continuity from pin L to pin K shall reappear.

4.6.32.3 Pitch servo alarm. Power as specified in 4.3.1.2.1 and 4.3.1.2.2 and the signal specified in 4.3.1.2.14 shall be applied. There shall be continuity from pins L to K of J202. The power of 4.3.1.2.2 shall be removed and the continuity between pins L and K shall disappear. The signal of 4.3.1.2.14 shall be removed and continuity between pins L and K shall reappear.

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4.6.33 Pitch amplifier. The unit shall be connected as specified in 4.3.1.2.1, 4.3.1.2.5, and 4.3.1.2.8. The output shall be measured between pins m and A of J201 and the phase reference voltage shall be from pin W to h of J201. With the unit connected as above, the following voltage shall exist:

Input Voltage	Output Voltage	Phase Relation
Connect pin W to pin h (0.0V rms)	2.5V rms max	- - - - - - -
0.060 ±0.002V rms	10.0 ±4V rms	0° ±20°
0.350 ±0.010V rms	14.0V min	0° ±20°

4.6.34 Leveling amplifier - null output voltage. The unit shall be connected as specified in 4.3.1.2.1 and an input signal applied as specified in 4.3.1.2.6 having the polarity and magnitude required to produce minimum rms output voltage across the load specified in 4.3.1.2.9. Null output shall not exceed 2V and the signal voltage required to obtain null output shall not exceed 0.020V dc.

4.6.34.1 The unit shall be connected as specified in 4.3.1.2.1, 4.3.1.2.6, 4.3.1.2.9, and 4.3.1.2.10. The output shall be measured between pins H and j of J201 and the phase reference voltage shall be from pins D to A of J201. With the unit connected as above, the following voltages shall exist:

Input Voltage	Input Polarity Pin i to Pin j		Output Voltage - Phase
0.050 ±0.002V dc	+	-	10.5 ±4V rms 180° ±30°
0.050 ±0.002V dc	-	+	10.5 ±4V rms 0° ±30°
0.250 ±0.005V dc	+	-	20.0 min 180° ±30°
0.250 ±0.005V dc	-	+	20.0 min 0° ±30°

The voltage between pins G and A of J201 shall be 3.5 ±0.5V rms.

4.6.35 Roll amplifier. The unit shall be connected as specified in 4.3.1.2.1, 4.3.1.2.7, and 4.3.1.2.11. The output shall be measured between pins q and A of J201 and the phase reference voltage shall be from pin f to e of J201. With the unit connected as specified, the following voltages shall exist:

Input Volts RMS	Output Volts RMS	Phase Relation
Connect pin f to pin e (0.0)	1 Max	- - - - - - -
0.10 ±0.002	21 ±7	180° ±30°
0.35 ±0.007	28 min	- - - - - - -

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4.6.36 Azimuth servo channel operation. Using the circuit shown on figure 8, power shall be applied and the test transmitter rotated through an angle of approximately 45° at a rate of approximately 45° per second. The azimuth servo shall move smoothly without noticeable hunting or overshooting.

4.6.37 Azimuth followup rate. The test transmitter shall be rotated at a rate between 30 and 40 rpm, first cw then ccw. The azimuth servo shall cause the output data transmitter signal on B202 to rotate through 360° in less than 2 seconds.

4.6.38 Azimuth slaving rate and cutout. With the flux valve, test transmitter, and B202 output at 0° heading, and mode selector switch S5 in the slaved position, the flux valve shall be rotated approximately 20° in an increasing heading direction. The output of B202 shall turn in an increasing heading direction at a rate between 1° and 2° per minute. When switch S2 is closed, the rotation of B202 shall stop. The same procedure shall be repeated, except the flux valve shall be rotated in a decreasing heading direction. The same requirement shall apply, except the output of B202 shall turn in a decreasing heading direction.

4.6.39 Latitude correction. With the BIAS-DEG/HR setting at 0, switch S5 in the DG position, and the tabulated voltage applied to terminals j to M of J901, the following tabulated B202 output rates shall exist (see figure 8):

S7 Position	Voltage Applied to j to M - J901	Correction Rate - °/hour
90° N	Voltage from V-U of J901	15 ±1 Incr Head
30° N	(.500 ±0.001) Voltage from V-U of J901	7.5 ±1 Incr Head
0°	0	0 ±1 Incr Head
90° S	Voltage from U-V of J901	15 ±1 Decr Head

4.6.40 Polarity. With switch S5 in the slaved position and the flux valve at 0° heading, switch S6 shall be momentarily energized. The flux valve shall be turned 30° in a numerically decreasing heading. The d-c voltage at J901-f shall be positive with respect to J901-g. The microampere meter needle in figure 8 shall point to the right and be in the "minus" sector of the meter. A rotation of S4 in a ccw direction shall cause the needle of the microampere meter to move to the left and the output of B202 shall be at 330° ±2° when the needle reaches its vertical position. This procedure shall be repeated, except the flux valve shall be rotated in a numercially increasing direction. The needle of the microampere meter shall point to left and be in the plus sector of the meter. A cw rotation of S4 shall cause the needle to move to the right and the output of B202 shall be at 30° ±2° when the needle reaches its vertical position.

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4.6.41 Annunciator. With the switch S5 in the slaved position and the output at B202 synchronized to the flux valve, the flux valve shall be rotated through an angle of ±0° 15', the microampere meter shall indicate a perceptible right, left movement with respect to its vertical position.

4.6.42 Deviation compensation. With switch S5 in the slaved mode, switch S2 in the cutout position, and B202 at 0°, the flux valve shall be set to 0° +3.5°. It shall be possible to obtain a null from f to g of J901 by adjusting the trim pot marked "N". The flux valve shall be set to 0° -3.5° and the trim pot marked "N" ajusted to obtain a null from f to g of J901. The above procedure shall be repeated at 15° increments through 345° and the trim pots marked "15" through "345", respectively, adjusted.

4.6.42.1 With the flux valve at 0°, S5 in the slaved position, and B202 at approximately 0° +2°, the output of B202 shall be allowed to approach 0° until slaving stops. The output shall be recorded. This procedure shall be repeated with B202 at approximately 0° -2°. The difference between the two final readings shall not exceed ±0° 30'. Neither of the final readings shall deviate from the flux valve setting by more than ±0°30'.

4.6.42.2 The procedure of 4.6.42.1 shall be repeated at 45° increments from 45° through 315°. The difference between the final readings at each flux valve setting shall not exceed ±0°30'. No final reading shall deviate from the flux valve setting by more than ±0° 30'.

Note: The deviation compensation trim pots may be adjusted to meet 4.6.42.1 and 4.6.42.2 as a part of only the initial individual tests or when sampling plan B tests are required, then as a part of the final sampling plan A tests.

4.6.43 Static settling. The readings obtained by following the procedure of 4.6.42.1 at 0°, 90°, 180° and 270° shall be such that the difference between the two readings at each of the four settings will not exceed ±0° 30'. The deviation trim pots shall not be used to meet this requirement.

4.6.44 Slaving accuracy. The readings obtained by following the procedure of 4.6.42.1 at 0°, 90°, 180°, and 270° shall be such that each of the 8 readings will be within ±0° 30' of their respective flux valve setting. The deviation trim pots shall not be used to meet this requirement.

4.6.45 Compass mode accuracy. With pins c, J and L of J901 connected to pin A of J901, B202 shall repeat the flux valve signal within ±2°. Rotation of the test transmitter shall not cause rotation of B202.

4.6.46 Azimuth servo accuracy. With the test transmitter and B202 at 0° and the mode selector switch S5 in the slaved position, the slaving cutout switch S2 shall be closed. The test transmitter shall be rotated through 360° in 90° increments without any overshooting in an increasing heading direction and in a decreasing heading direction. The output of B202 shall be recorded for each position. The difference in readings for the same heading for decreasing and increasing heading directions shall not exceed 0° 20'. The difference between the test transmitter and B202 output shall not exceed 0° 30'.

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4.6.47 Bias correction. With S5 in the DG position, the following tabulated changes in the correction rate shall result by changing the BIAS-DEG/HR setting from 0 as shown.

Bias-Deg/Hr Setting	Change in Correction Rate (Deg/Hr)
+1	Increase 1 ±0.2
+2	Increase 2 ±0.2
-1	Decrease 1 ±0.2
-2	Decrease 2 ±0.2

4.6.48 Automatic gain control. The flux valve shall be placed in a magnetic field with a horizontal component of 0.08 oersteds. The compass adapter shall meet the requirements of 4.6.38 and 4.6.44. This procedure shall be repeated except the horizontal component shall be 0.40 oersteds.

4.6.49 Dynamic accuracy. With the compass adapter connected in accordance with figure 8, operating in the slaved mode and synchronized, the flux valve specified in 4.3.1.5.3 shall be oscillated through ±5° either side of null at 6 cpm. The output of B202 shall be read each minute after start of the oscillation until 3 consecutive readings are within 0.1° of each other; this shall occur in less than 45 minutes from the start of the test. The difference between the initial null value and the final reading of 3 consecutive readings shall not exceed 0.75°.

4.6.50 Manual synchronization. With the flux valve at 180° and B202 at 0°, the compass adapter shall be manually synchronized in an increasing heading direction by rotating S4. It shall take not more than 4.5 seconds to reach synchronization. The above procedure shall be repeated, except in the decreasing heading direction.

4.6.51 Switchover synchronization. With the flux valve at 0°, B202 at 170° and the switch S5 in the slaved position, S5 shall be momentarily switched to the DG position and back to the slaved position. The output of B202 shall automatically rotate to 0° ±2° within 4.5 seconds. The procedure shall be repeated, except B202 shall be set to 190°.

4.6.52 Push to sync operation. With the flux valve at 0°, B202 at 170°, and switch S5 in the slaved position, switch S6 shall be momentarily energized. The output of B202 shall automatically rotate to 0° ±2° within 4.5 seconds. The procedure shall be repeated, except B202 shall be set to 190°.

4.6.53 Heading set slew. Switch S5 shall be in the DG position and the outputs of data transmitter B202 and differential synchro B208 shall each be connected to a heading repeater. Switch S4 shall be depressed and rotated cw; the heading repeaters shall show an increasing heading. The position of switch S4 shall be varied and the rate of heading change shall vary approximately proportionally with the position of switch S4. The maximum rate shall be not less than 50° per second. The minimum rate shall average less than 3° per second when measured over 30-second periods. This procedure shall be repeated, except the rotations shall be in a ccw direction.

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4.6.54 Autopilot slewing decoupling. Continuity between pins F and G of J901 shall be transferred to between pins G and H of J901 during the operations specified in 4.6.45, 4.6.50, 4.6.51, 4.6.52, 4.6.53, and 4.6.62.1.

4.6.55 Synchro zeroing and sensing. With B202 at 0°, the test transmitter at 0° ±0.5°, and slaving cut out, the output data transmitters shall read as follows. The output data transmitters shall follow the rotation of the test transmitter in both directions.

Transmitter	Position of S3	Output
B203	4	0° ±0° 20'
B204	5	0° ±0° 20'
B205	2	0° ±0° 20'
B206	1	0° ±0° 40'
B202	3	N/A

4.6.56 Synchronizing indicator. With switch S805 in the slaved position, a current of 50 ±5 ua dc shall be applied to pins a (+) and b (-) of J801. The pointer shall deflect to the minus scale markings.

4.6.57 Mode selector switch. The specified resistance shall exist between the following connector pins for the two switch positions of S805.

Connector Pins	Slaved Mode	DG Mode
Z - Y	Less than 1 ohm	30 megohms (min)
Z - f	30 megohms (min)	Less than 1 ohm
h - g	Less than 1 ohm	30 megohms (min)
K - L	30 megohms (min)	Less than 1 ohm
L - X	Less than 1 ohm	30 megohms (min)

4.6.58 Set heading control. The specified resistance shall exist between the following connector pins for the two switch positions. The knob shall not be rotated and the mode selector switch shall be in the slaved position.

Connector Pins	Normal	Depressed
Z - Y	Less than 1 ohm	30 megohms (min)
U - T	30 megohms (min)	500 ±50 ohms
R - S	Less than 1 ohm	30 megohms (min)
c - d	Less than 1 ohm	30 megohms (min)

When the knob is depressed, rotation to full cw position shall increase the resistance between pins T and U to 1,000 ±100 ohms. Rotation to full ccw position shall decrease the resistance between pins T and U to 30 ohms, or less.

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4.6.59 Latitude control. With hemisphere switch S807 and the latitude control knob in the following specified positions, the resistance shown shall exist between the indicated pins of J801.

Hemisphere Switch	Latitude Knob	Resistance	Connector Pins
S	Any	Less than 1 ohm	P & M
N	Any	120 - 135 ohms	P & M
N	Any	Less than 1 ohm	N & M
S	Any	120 - 135 ohms	N & M

With S807 in the south position, S805 in the DG position, and the latitude control dial at 0°, the resistance between pins P and h shall be less than 1 ohm. Rotating the latitude control dial from 0° thru 90° shall result in an increasing resistance to between 120 and 135 ohms, without interruption, between pins P and h.

4.6.60 Push to sync. A resistance of less than 1 ohm between pins W and X of J801 shall exist in the depressed position of the push-to-sync knob. This resistance shall be infinite with the knob in normal position (nondepressed).

4.6.61 Lighting. A voltage of 28 ±0.5V dc shall be applied to pins A and j (+) of J801 and the panel shall light up.

4.6.62 System compatibility

4.6.62.1 Power application synchronization. The control assembly shall be connected in accordance with figure 9 with no power applied. With switch S805 in the slaved position, power shall be applied. The output at B202 shall slew to the input from the flux valve within ±2°.

4.6.62.2 Power consumption. The control assembly shall be connected as shown on figure 9 without the application of power. Power consumption shall be checked under starting and running conditions and shall not exceed 350 va starting and 200 va running.

4.6.62.3 Special third-gimbal operation. The control assembly shall be connected to a DCK-10/A24G-1 third gimbal controller conforming to MIL-C-26601. The control assembly shall be operating in the DG mode. The displacement gyroscope assembly shall be mounted on a fixture permitting 360° rotation in roll, pitch, and azimuth. The following tests shall be met:

4.6.62.3.1 Sustained vertical position. The displacement gyroscope assembly shall be rotated manually through 360° in pitch and the output of the pitch synchro (J361 - a, b, c) shall indicate a continuous 360° pitch rotation. From the level position, the displacement gyroscope assembly shall be rotated 90° ±3° in pitch. In this attitude, the azimuth axis of the fixture shall be so rotated that a 360° roll motion will be applied to the displacement gyroscope assembly. The roll synchro (J361 U, V, W) shall

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indicate a roll motion equivalent to the actual displacement in roll applied to the displacement gyroscope assembly. The displacement gyroscope assembly shall be leveled and then rotated in pitch through the vertical to 100° ±5° in pitch. The heading synchro J902 - a(x), b(y), c(z), shall be monitored and as the displacement gyroscope assembly passes through the vertical position, a 180° change in heading shall be noted. With the displacement gyroscope assembly in the vertical position, the assembly shall be slowly rotated about its azimuth axis (yaw) and the roll (J361 U, V, W) and pitch (J361 - a, b, c) outputs monitored. Between 15° to 22° of yaw motion, the roll signal shall suddenly drive 90° ±5° and the pitch signal shall change approximately 20°.

4.6.62.3.2 Normal operation. The displacement gyroscope shall be operated for a period of 15 minutes on a roll, pitch, and yaw mechanism, moving at 5 to 7 oscillations per minute with a total amplitude of 15° in each of the 3 axes. The roll, pitch, and yaw mechanism shall be leveled within ±1/10° and the control assembly attitude indication shall be read immediately. The vertical gyro shall be within 1/2° of the local gravity vertical as indicated by the output synchro signals.

4.6.62.4 Static settling test. The control assembly shall be mounted as specified in 4.3.3b, connected as shown on figure 9, operated in the slaved compass mode, and set to a synchronized null. The synchro repeater specified in 4.3.1.5.2 shall be connected to J902 a(x), b(y), and c(z). The displacement gyroscope assembly shall be rotated in azimuth 10° from the null position. The synchro repeater shall be read after a period of 15 minutes. This procedure shall be repeated for each side of null and the difference in the two final setting positions shall be not greater than 1/2°.

4.6.63 Case leakage. The sealed cases shall be tested for leaks by use of a mass spectrometer-type helium leak detector or other means approved by the procuring activity. The leak rate shall not be sufficient to cause more than 10 percent loss or contamination of the filling medium after 1,000 hours at approximately 1 atmosphere differential pressure.

4.6.64 Voltage and frequency variation. The input voltage to each component of the control assembly shall be reduced to 103V and 380 Hz. Components shall meet the tests specified in 4.6.12, 4.6.13, 4.6.19, 4.6.36, 4.6.37, 4.6.39, and 4.6.49, as applicable, with a 10 percent increase in tolerances and 4.6.21 and 4.6.23.1 with a 25 percent increase in tolerances.

4.6.65 Low temperature operation. The components shall be properly connected without power applied and placed in a chamber at an ambient temperature of -54° ±2° C for a period of 4 hours. At the end of this period and while at the reduced temperature, the components shall be subjected to the tests specified in 4.6.2, 4.6.3, 4.6.4, 4.6.10, 4.6.12, 4.6.13, 4.6.19, 4.6.21, 4.6.22, 4.6.23, 4.6.27, 4.6.36, 4.6.37, 4.6.38, 4.6.39, 4.6.44, 4.6.49, and 4.6.56 through 4.6.61, as applicable, except as follows:

4.6.65.1 Initial erection. Within 1 minute after application of power as specified in 4.3.1.1.22.2, the pitch output shall not exceed 1° and the roll shall not exceed 1.5°.

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4.6.65.2 Verticality. A minimum of 5 minutes after application of power as specified in 4.3.1.1.22, the verticality requirements shall be met.

4.6.65.3 Latitude correction gyroscope. The error shall not exceed 3.5° per hour.

4.6.65.4 Final erection. Within 5 minutes after completion of the procedure specified in 4.3.1.1.22.2, the final erection requirements shall be met.

4.6.65.5 Slaving accuracy. The error shall not exceed ±0° 45 .

4.6.66 High temperature. The components shall be properly connected as a control assembly with no power applied and subjected to a high temperature test in accordance with MIL-STD-810, method 501, procedure II with the highest operating temperature 98°C. While at the high temperature, power shall be applied and the gyro shall be rotated through 360° about the pitch, roll, and azimuth axes. All components shall perform their respective functions without binding or erratic movements as determined by observing the heading outputs shown on figure 8, pitch outputs of J361, pins d, e, and f, and roll outputs on J361, pins g, h, and j. After the components have returned to room temperature, they shall meet the tests specified in 4.6.2, 4.6.3, 4.6.4, 4.6.10, 4.6.12, 4.6.13, 4.6.21, 4.6.22, 4.6.23, 4.6.27, 4.6.37, 4.6.38, 4.6.39, 4.6.49, and 4.6.56 through 4.6.61, as applicable.

4.6.67 Dielectric strength. Each isolated circuit, except those connected to the electrolytic switches and leveling pickoff, shall have a potential of 500V dc applied between it and all other such circuits and the gyro case for a period of 10 seconds. They shall exhibit a resistance of not less than 50 megohms during application of voltage.

4.6.68 Vibration error. A complex wave vibration test shall be conducted to determine the ability of the equipment to function in a vibration environment. During the vibration test, the input vibration to the equipment shall have a power spectral density of 0.015 g^2/Hz for each frequency from 20 to 2,000 Hz. In order to accomplish this representation, the loading effect of the unit under test shall be eliminated by proper load compensation. The random vibration shall be applied to the equipment along the three mutually perpendicular axes of the system components. The signal source shall be either a prepared magnetic tape or a noise generator. The duration shall be from 10 to 15 minutes to properly observe the errors. The components may be individually vibrated for this test.

4.6.68.1 Displacement gyroscope assembly. The gyroscope assembly shall be connected in accordance with 4.3.1.1.1 through 4.3.1.1.13 and vibrated. The input vibration to the gyroscope assembly shall have a power spectral density of 0.002 g^2/Hz for each frequency from 20 to 2,000 Hz. The pitch output signals at pins J361-a, b, c, and

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J361-d, e, and f, the roll output signals at pins J361-X, Y, Z, J361-U, V, W, and J361-g, h, j shall not vary by more than 1° and shall return to the initial position when vibration ceases. The azimuth output signals at pins J361-R, S, T and J361-M, N, P shall not have changed by more than 1° with proper correction for the earth's rotation.

4.6.68.2 Amplifier-power supply. The amplifier-power supply shall be vibrated in accordance with 4.6.68 and shall meet the tests indicated in 4.5.1.2.

4.6.68.3 Compass adapter. With the compass adapter slaved to a flux-valve setting of 0°, the vibration specified in 4.6.68 shall be applied. The output of B202 shall not exceed 0° ±1°.

4.6.68.4 Compass controller. The compass controller shall be mounted in its normal operating position, connected in accordance with 4.3.1.4, and vibrated in a plane inclined 45° to the horizontal plane in a circular motion with a diameter between 0.009 and 0.011 inch. The frequency of applied vibration shall be varied slowly from 5 to 50 Hz. The controls on the compass controller shall not move in relation to the panel and the annunciator needle shall not vibrate excessively. The compass controller shall then meet the tests specified in 4.6.56 through 4.6.61.

4.6.69 Low field strength. The compass adapter shall be connected properly, operating with the circuit on a synchronized null, and subjected to the dynamic accuracy test specified in 4.6.49, except that the oscillating signal shall be equivalent to a magnetic detector oscillating about the reference null setting in a horizontal magnetic field of 0.080 ±0.002 oersted and a vertical field of 0.860 ±0.002 oersted. The resulting shift in synchronized null shall not exceed 0.75°. This test shall be run on three headings not less than 60° apart.

4.6.70 Balance. With power applied as necessary, the azimuth axis maintained in a horizontal position, and the directional gyro spin-axis vertical, the drift of the directional gyro shall not exceed 2-1/2° in 1/2 hour after proper correction for the earth's rotation when the directional gyro pitch gimbal is oscillated through approximately ±2° about the azimuth axis at a rate of 5 to 7 cpm.

4.6.71 Radio noise interference and susceptibility. The radio noise interference and susceptibility test shall be conducted in accordance with MIL-I-6181 over a frequency range of 150 KHz to 400 MHz for interference and over a frequency range of 150 KHz to 1,000 MHz for susceptibility. The control assembly shall be connected properly and operating on a 15° roll, pitch, and yaw mechanism.

4.6.72 Acceleration. With proper power applied, the components of the control assembly shall be accelerated first along their vertical axes and then along each of two axes that are perpendicular to the vertical axis and to each other. The components shall be accelerated in each direction in each of the above axes for a period of 1 minute. During the acceleration period, the components shall operate satisfactorily. At the end of this acceleration, the components shall meet their respective individual tests. The components may be accelerated individually.

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4.6.72.1 With the exception of the displacement gyroscope, the components shall be subjected to an acceleration of 20g.

4.6.72.2 The displacement gyroscope shall be subjected to the following g-levels:

Fore	Aft	Up	Down	Lateral
6.1	6.1	3.3	9.0	6.1

4.6.73 Temperature-altitude. The components shall be properly connected as a control assembly with no power applied and subjected to a temperature-altitude test in accordance with MIL-STD-810, method 504, procedure I, class 2 equipment, except the altitude shall be 80,000 feet. At the temperature extremes, the time required for initial erection of the gyros and for the system to synchronize shall not exceed 5 minutes. The gyro shall be rotated through 360° about the pitch, roll, and azimuth axes and all components of the control assembly shall perform their respective functions without binding or erratic movements as determined by observing the heading outputs shown on figure 8, pitch outputs of J361, pins d, e, and f, and roll outputs of J361, pins g, h, and j. After the components have returned to room temperature, they shall meet the tests specified in 4.6.2, 4.6.3, 4.6.4, 4.6.10, 4.6.12, 4.6.13, 4.6.21, 4.6.22, 4.6.23, 4.6.27, 4.6.37, 4.6.38, 4.6.39, 4.6.49, and 4.6.56 through 4.6.61, as applicable.

4.6.74 Temperature shock. The control assembly components shall be subjected to a temperature shock test in accordance with MIL-STD-810, method 503, procedure I.

4.6.75 Shock. The control assembly components shall be subjected to a shock test in accordance with MIL-STD-810, method 516, procedures I and III. The shock pulses shall be in accordance with figure 516-2. After completion of procedure I testing, the components shall meet the applicable individual tests.

4.6.76 Humidity. The components shall be subjected to a humidity test in accordance with MIL-STD-810, method 507, procedure I after which they shall meet the tests specified in 4.6.27 through 4.6.41, 4.6.43 through 4.6.61, and 4.6.63, as applicable. There shall be no evidence of detrimental corrosion or rusting of external parts.

4.6.77 Fungus. The control assembly components shall be subjected to a fungus test in accordance with MIL-STD-810, method 508, procedure I after which they shall meet the tests specified in 4.6.27 through 4.6.41, 4.6.43 through 4.6.61, and 4.6.63, as applicable. There shall be no evidence of detrimental deterioration, corrosion, or fungus growth.

4.6.78 Rain. The nonhermetically sealed components shall be subjected to a rain test in accordance with MIL-STD-810, method 506, procedure I after which they shall meet the tests specified in 4.6.27 through 4.6.41 and 4.6.43 through 4.6.61, as applicable. There shall be no evidence of water penetration or damage to the components.

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4.6.79 Salt fog. The components shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509, procedure I after which they shall meet the tests specified in 4.6.27 through 4.6.41, 4.6.43 through 4.6.61, and 4.6.63, as applicable.

4.6.80 Dust. The nonhermetically sealed components shall be subjected to a dust test in accordance with MIL-STD-810, method 510, procedure I. Immediately after reaching stabilization in step 2, the second 6-hour test at 63°C shall be performed. At the conclusion of this test, the components shall meet the tests specified in 4.6.27 through 4.6.41 and 4.6.43 through 4.6.61, as applicable. There shall be no evidence of dust penetration or damage to the components.

4.6.81 Explosive atmosphere. The nonhermetically components shall be subjected to an explosive atmosphere test in accordance with MIL-STD-810, method 511, procedure I.

4.6.82 Acoustical noise. The components shall be properly connected, operated, and subjected to an acoustical noise test in accordance with MIL-STD-810, method 515, procedure I, category B. The pitch and roll outputs shall be monitored throughout the test and shall not vary by more than 1/2° from the initial position established at 10 percent of the noise level. When the noise level is reduced to 10 percent, the pitch and roll outputs shall return to the initial position.

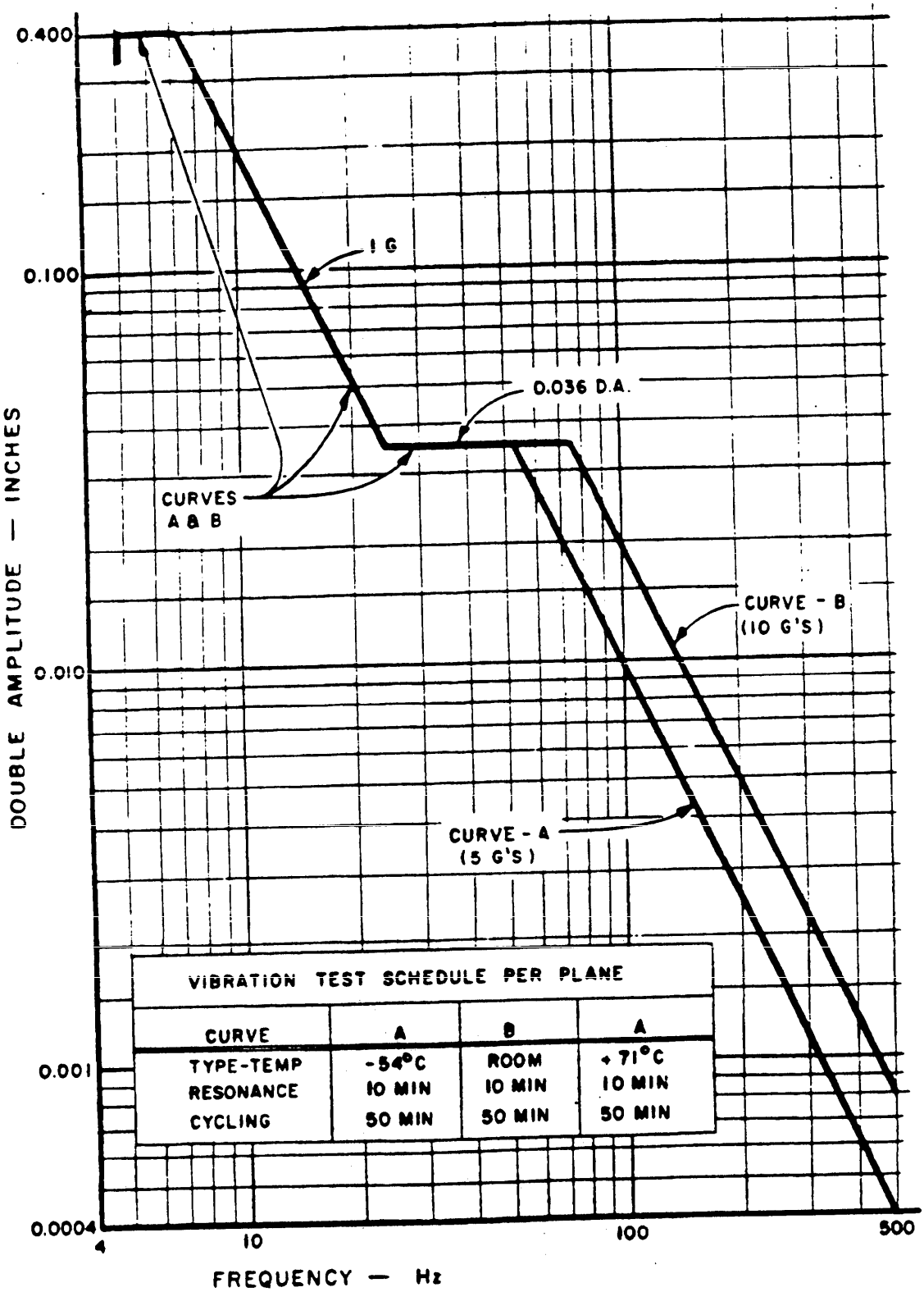
4.6.83 Sealing thermal shock. The hermetically sealed components shall be subjected to eight cycles of alternate immersions in tap water maintained at 85° ±5°C and 5° ±4°C. The length of time for each immersion shall be 30 minutes with not more than 5 seconds between immersions. After completion of the immersions, the components shall meet the test specified in 4.6.63. No damage to the hermetic seal shall result from this test.

4.6.84 Vibration failure. The components shall be subjected to the following tests:

4.6.84.1 Compass controller. The compass controller shall be mounted on the vibration apparatus in its normal operating position. While being operated, the instrument shall be vibrated with a circular motion of 0.018 to 0.020 inch diameter in a plane inclined 45° to the horizontal plane and the frequency of vibration shall be varied uniformly from 5 to 50 Hz and return once each hour for a 3-hour period.

4.6.84.2 Displacement gyroscope assembly. The displacement gyroscope assembly shall be subjected to a vibration failure test in accordance with MIL-STD-810, method 514, equipment category (b), curve C, except as modified herein on figure 13, curve A. The displacement gyroscope assembly shall be properly connected with power applied as specified in 4.3.1.1.1 through 4.31.1.13. During vibration, the gyros and gimbals shall be in approximately the correct position. At the completion of vibration, the gyroscope shall meet the tests specified in 4.6.2, 4.6.10, 4.6.12, 4.6.13, 4.6.21, and 4.6.23.

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VIBRATION TEST SCHEDULE PER PLANE

CURVE	A	B	A
TYPE-TEMP	-54°C	ROOM	+71°C
RESONANCE	10 MIN	10 MIN	10 MIN
CYCLING	50 MIN	50 MIN	50 MIN

FIGURE 13. Vibration

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4.6.84.3 Compass adapter. The compass shall be subjected to a vibration test in accordance with MIL-STD-810, method 514, equipment category (b) curve C, except as modified herein on figure 13, curve A. Power shall be applied as specified in 4.3.1.3. At completion of this test, the compass adapter shall meet the tests specified in 4.6.1, 4.6.36 through 4.6.41, and 4.6.43 through 4.6.54 except the slaving error shall not exceed ±0° 45′.

4.6.84.4 Amplifier-power supply. The amplifier-power supply shall be subjected to a vibration test in accordance with MIL-STD-810, method 514, equipment category (b), curve C, except as modified herein on figure 13, curve A. Power shall be applied as specified in 4.6.86.2.1.2. At the completion of this test, the amplifier-power supply shall meet the applicable individual tests.

4.6.85 Vibration endurance. The components of the control assembly except the compass controller shall be subjected to a vibration endurance test in accordance with MIL-STD-810, method 514, equipment category (b), curve D, except as modified herein on figure 13, curve B. Power shall be applied as specified in 4.6.84, as applicable. At the completion of this test, the components shall meet the tests specified in 4.6.1, 4.6.3, 4.6.4, 4.6.23, 4.6.27, 4.6.33 through 4.6.35, 4.6.37 through 4.6.39, 4.6.44, and 4.6.50, as applicable, except as follows:

4.6.85.1 Final erection. The error in either roll or pitch shall not exceed ±1° 30′.

4.6.85.2 Verticality. The change shall not exceed ±2°.

4.6.85.3 Directional gyro drift. The drift rate shall not exceed ±5° per hour for the 1/2-hour test period.

4.6.85.4 Azimuth follow rate. The time necessary for the output of B202 to rotate through 360° shall not exceed 4 seconds in either direction.

4.6.85.5 Azimuth slaving rate and cutout. The minimum slaving rate shall be 1/2° per minute.

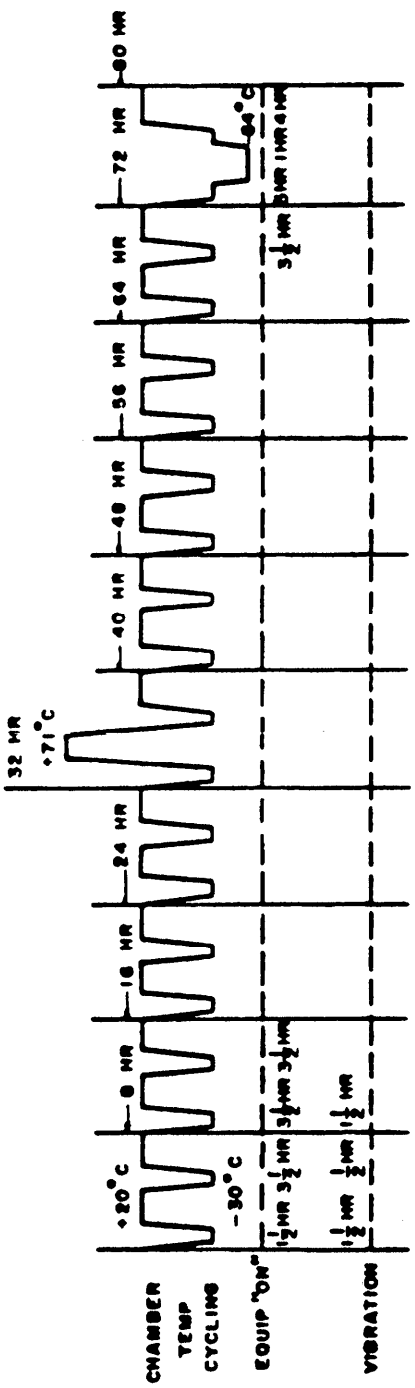
4.6.85.6 Latitude correction gyroscope. The test shall be limited to the 60° N setting and the error shall not exceed ±2° per hour.

4.6.85.7 Slaving accuracy. The error shall not exceed ±1°.

4.6.86 Reliability

4.6.86.1 Preproduction. The four systems specified in 4.4.1.1 shall be properly connected and operated to assure proper set up. The systems, less compass controllers, shall be placed in test chambers which shall meet the environmental conditions specified on figure 14. During the ON cycle, the system shall be exercised by a 3-axis, 15° total movement, 5-to-7 cpm, reversing scorsby action applied to the displacement gyroscope and by the vibration shown on figure 14. The attitude of the gyro shall

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NOTES:

a. Environment cycle repeats every 80 hours.

b. Temp rate of change of chamber shall be 0.1°C - 0.5°C per second.

c. Vibration level 0.2g at 20 to 30 Hz.

d. Off periods for equipment ON and OFF are $\frac{1}{2}$ hour (except 76th hour).

e. Monitoring of system outputs for catastrophic type failures performed during each ON cycle.

f. Satisfactory performance tests on systems and individual units conducted every 160 hours of environment cycle to detect performance - deviation type failures.

FIGURE 14. Reliability Test Environment Cycles

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be indicated within ±2° by the outputs of the pitch and roll synchros of the gyro at pins d, e, and f, and X, Y, and Z of J361. The output of B202 on pins a, b, and c of J902 shall repeat the input to B207 within ±2° when the system is in the slaved mode and synchronized. After each 160-hour cycle of operation, the equipment shall meet the following. Rejection shall be in accordance with 4.4.1.1.

a. With the components properly connected as a system and with the gyroscope level within ±0° 6′, power shall be applied and the following checks made:

(1) Time - 0 to 12 seconds - The voltage from pin c to A of J201 shall be approximately zero. At 12 ±6 seconds, the voltage shall increase to approximately 50V.

(2) Time - 60 +20 -8 seconds - The voltage from pin c to A of J201 shall decrease to approximately 15V.

(3) Pitch - The output at terminals d, e, and f of J361 shall not exceed the equivalent of ±1.5° at the voltage reduction specified in a(2).

(4) Roll - The output at pins X, Y, and Z of J361 shall not exceed the equivalent of ±1.5° immediately after the operation specified in a(2).

(5) Heading - The output at pins a, b, and c of J902 shall repeat the input to r, s, and t of J902 within ±1.5° at the voltage reduction specified in a(2) with the DG-SLAVED switch in the SLAVED position.

b. Gyro operation - The outputs at pins d, e, and f of J361, at pins X, Y, and Z of J361, and at pins a, b, and c of J902 shall follow smoothly when the gyro is rotated through 360° in both directions in each of the three perpendicular axes.

c. Slaving - With the system operating in the slaved mode and synchronized, the flux valve shall be rotated approximately 30° in an increasing heading direction. The output of pins a, b, and c of J902 shall rotate in an increasing heading direction at a rate of 1.5° ±0.50 per minute. This procedure shall be repeated, except in a decreasing heading direction. The output shall rotate in a decreasing direction at a rate of 1.5 ±0.50 per minute.

d. Erection cutout - Application of 115V ac, 400 Hz to pins M and N of J202 shall cause the voltage at pins R and c, respectively, of J201 to be reduced to zero.

e. Slaving cutout - With the application of 115V ac, 400 Hz to pin k of J901, the slaving action described in procedure c shall cease.

f. Free mode operation - With the DG-SLAVED switch in the DG position, the output at pins a, b, and c of J902 shall rotate at a rate not exceeding ±2° per hour after correction for the mean drift of the gyro and the local latitude. The gyroscope shall be mounted in accordance with 4.3.3d for this test.

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g. Roll operation - The output at pins X, Y, and Z of J361 shall repeat the attitude of the gyroscope within 1.5°.

h. Pitch operation - The output at pin W of J362 shall not exceed 3V with the gyro mounted in accordance with 4.3.3d.

i. Synchronization - With the DG-SLAVED switch in the SLAVED position and the output at pins a, b, and c of J902 repeating the input at pins r, s, and t of J902, the input at pins r, s, and t of J902 shall be rotated 30° in an increasing heading direction. The PUSH-TO-SYNCH switch shall be depressed momentarily and the output of B202 at pins a, b, and c shall slew to the input at pins r, s, and t within ±2°.

j. Output synchros - With the DG-SLAVED switch in the DG position, the gyro shall be rotated through 360° in each direction. The output at pins f, g, and h, k, m, and n and V, W, and X of J902 shall follow smoothly.

k. Power consumption - With the system operating properly, the power consumption shall not exceed 200 va running.

4.6.86.1.1 Longevity. Three of the four control assemblies specified in 4.6.86.1 shall continue to be tested through a longevity test for a total of 2,000 hours each. The control assemblies shall be cycled as follows: 22 hours operating and 2 hours nonoperating with each component meeting its respective individual tests every 17 cycles. Where applicable, the elapsed time indicators shall be used to record the test time. Certified time records shall be maintained on the compass controller. Upon completion of this test, the components shall meet their respective sampling plan A tests.

4.6.86.1.1.1 Displacement gyroscope assembly. The gyroscope assembly shall have power applied in accordance with 4.3.1.1.1 through 4.3.1.1.13 and shall be operated on a 15° total movement roll, pitch, and yaw mechanism having 5 to 7 oscillations per minute.

4.6.86.1.1.2 Amplifier-power supply. The amplifier-power supply shall be connected in accordance with 4.3.1.2.1, 4.3.1.2.5 through 4.3.1.2.7, and 4.3.1.2.8 through 4.3.1.2.11. The input voltages shall be approximately as follows:

a. 4.3.1.2.5 0.060V

b. 4.3.1.2.6 0.050V

c. 4.3.1.2.7 0.100V

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4.6.86.1.1.3 Compass adapter. The compass adapter shall be connected in accordance with figure 8, switch S5 in the SLAVED position, and the test transmitter shall be oscillated through 15° heading at a rate of 5 to 7 cpm.

4.6.86.1.1.4 Compass controller. The compass controller shall be connected in accordance with 4.3.1.4.3 and all switches and dials operated twice at 24-hour intervals.

4.6.86.2 Production. The three control assemblies specified in 4.5.2.3 shall be subjected to a production reliability test in accordance with 4.6.86.1 except for a total of 1,800 hours. Rejection shall be as specified in 4.5.2.3.1.

4.6.86.3 Failure. Where a condition is specified as a requirement, a failure will be considered as any electrical or mechanical condition which fails to meet the requirements specified herein.

4.6.86.4 Test records. Throughout the tests, the contractor shall maintain adequate test records. The records shall be prepared in a form suitable for reproduction. As a minimum, a test data log, operator's log, equipment failure log, and a failure diagnosis and corrective action report shall be kept. The forms shall contain the data specified in MIL-STD-781 (minimum). Reproducible copies of the data shall be submitted to the procuring activity within 30 days after the completion of each test phase.

4.6.86.5 Disposition of failed units. Disposition of failed units shall be as specified by the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified by the procuring activity, the control assembly shall be preserved and packaged in accordance with MIL-P-116, method II for nonhermetically sealed components and method III for hermetically sealed components. Each unit shall be packed and prepared for shipment in accordance with MIL-STD-794. The level shall be as specified by the procuring activity.

5.2 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be as follows, as applicable.

Control Assembly, Attitude, Gyroscopic AF/A24G-1A
Adapter, Compass ADK-182A/A24G-1A
Gyroscope Assembly, Displacement SBK-8/A24G-1A
Amplifier-Power Supply ASK-12A/A24G-1A
Controller, Compass DCK-100/A24G-1A

5.2.1 Reinspection date. Reinspection date markings shall be as specified by the procuring activity.

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6. NOTES

6.1 Intended use. The AF/A24G-1A control assembly covered by this specification is intended for use in aircraft to provide pitch, roll, and heading displacement signals throughout all maneuvers of the aircraft for indication and tie-in to other equipment. Provisions are made for various types of operation and error compensation, depending upon the application and associated equipment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. If mounting base is required (see 3.9.4 and 3.10.12)

c. When sampling plan B and C tests will not be conducted

d. Levels of packaging and packing desired and reinspection date markings

e. All failures occurring during testing to be reported to the procuring activity within 2 working days.

6.4 Marginal indicia. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 11

Review activity:
Air Force - 71

Preparing activity:
Air Force - 11

Project No. 6615-F002

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

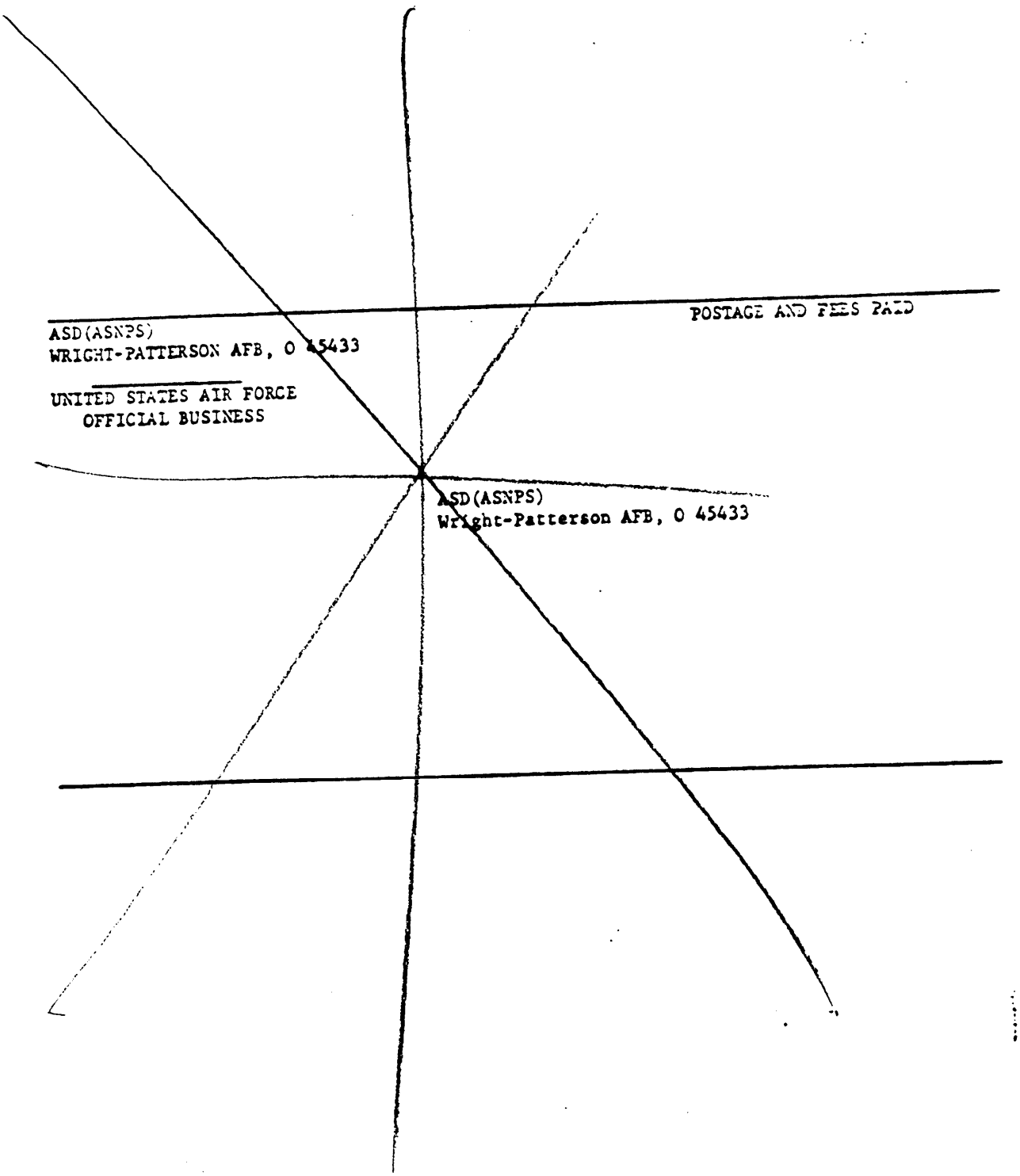
DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC-WPAFB-OCT 67 2M

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